UNIT 3. The Memory System

Overview

- Basic memory circuits
- Organization of the main memory
- Cache memory concept
- Virtual memory mechanism

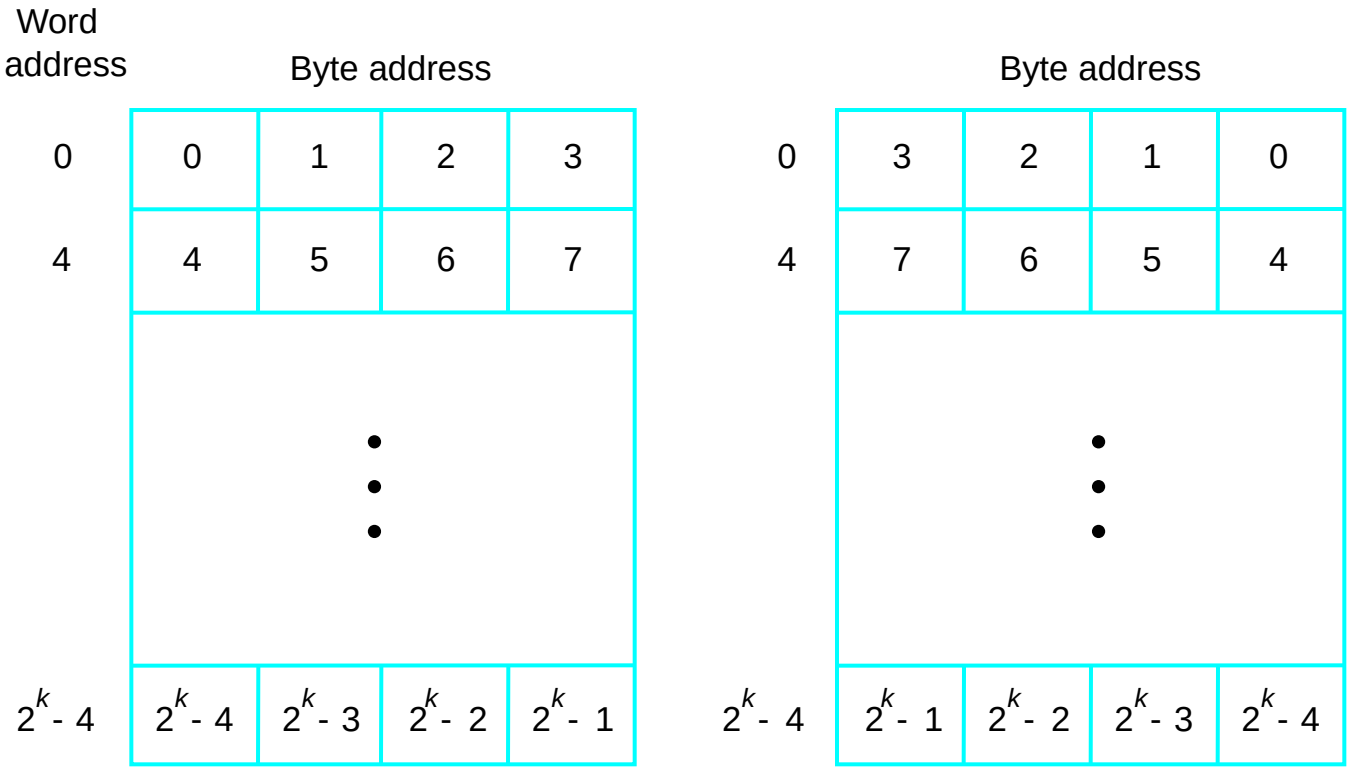
Some Basic Concepts

Basic Concepts

- The maximum size of the memory that can be used in any computer is determined by the addressing scheme.

  16-bit addresses = 2^{16} = 64K memory locations

- Most modern computers are byte addressable.



Traditional Architecture

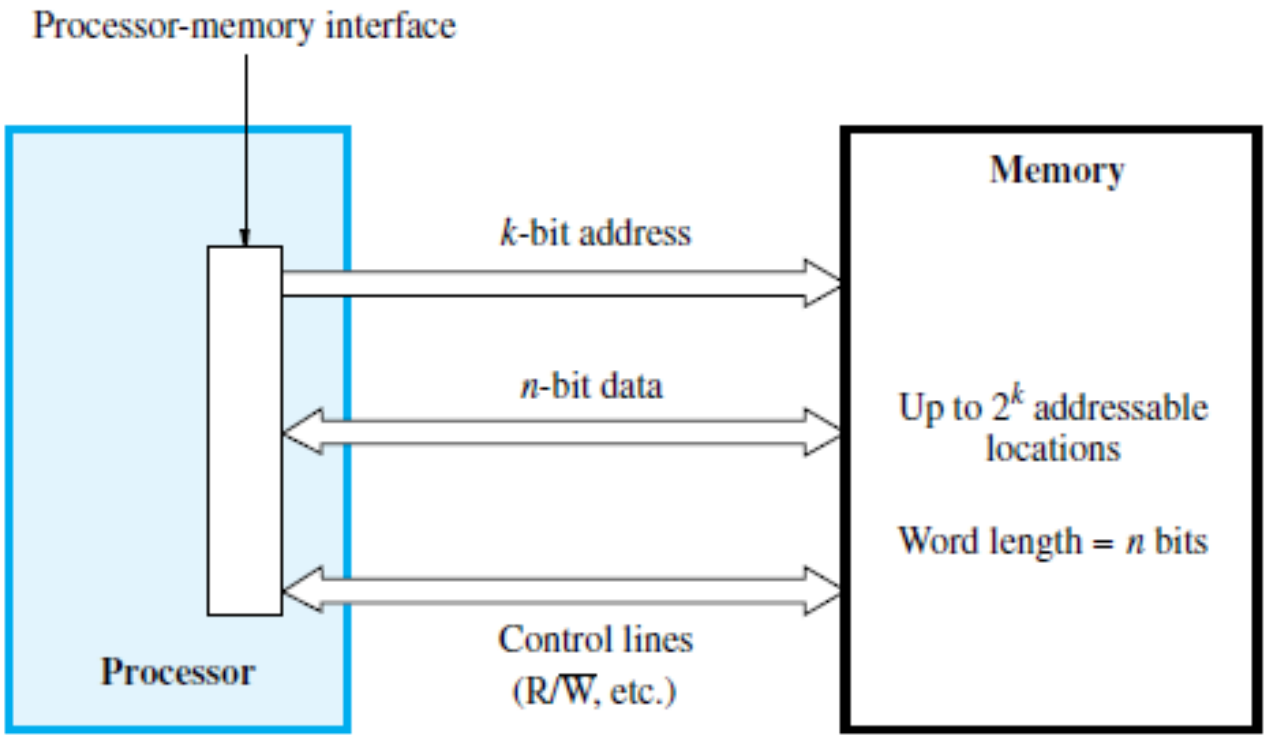


Figure 8.1 Connection of the memory to the processor.

Some basic concepts(Contd.,)

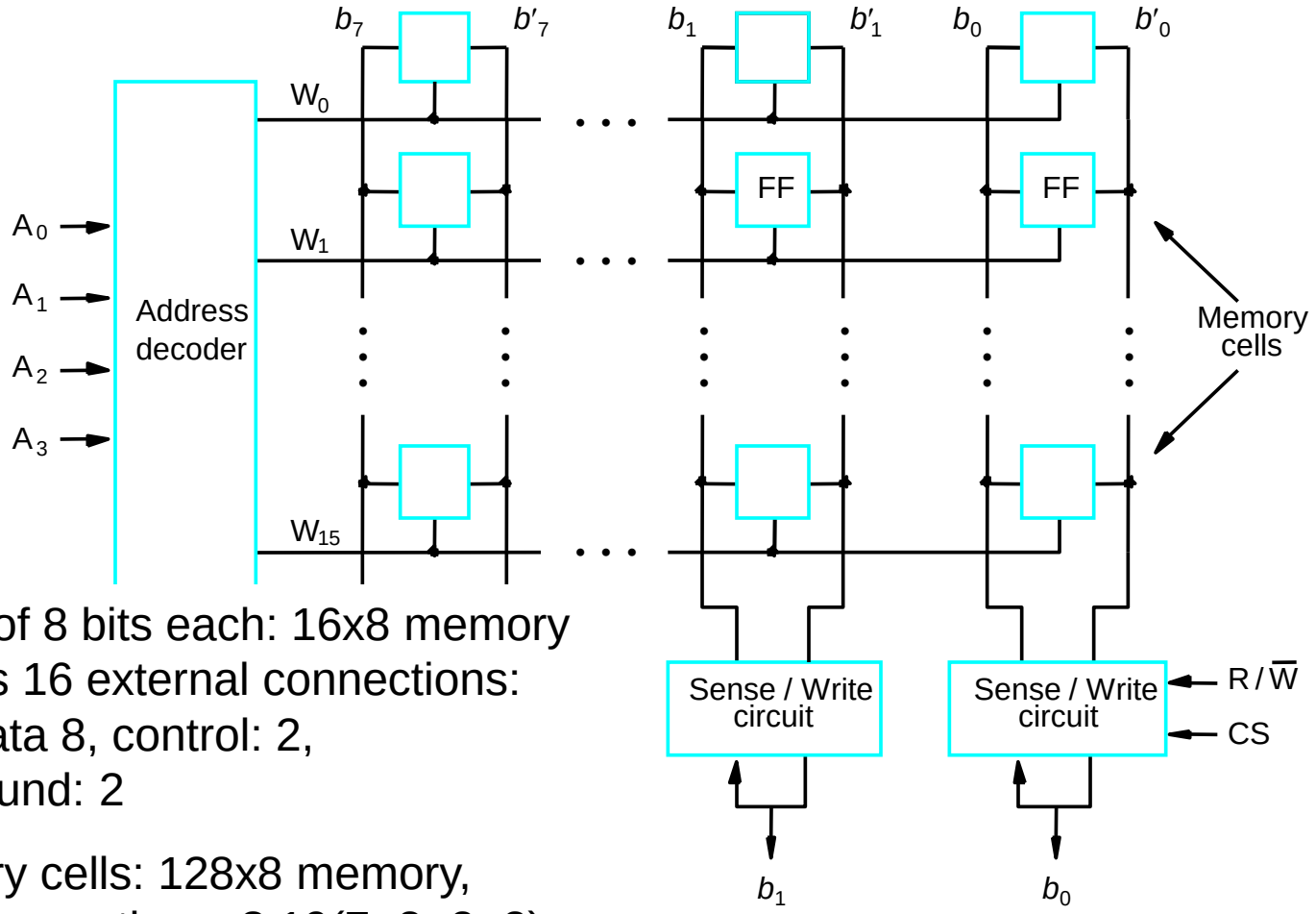
- Measures for the speed of a memory:
  - memory access time: time interval between the initiation of memory data transfer and the completion of the operation
  - memory cycle time: minimum time delay required between the initiation of two successive memory operations. Eg: time between two successive Read operarions
- An important design issue is to provide a computer system with as large and fast a memory as possible, within a given cost target.
- Several techniques to increase the effective size and speed of the memory:
  - Cache memory (to increase the effective speed).
  - Virtual memory (to increase the effective size).
- “Block transfer” – bulk data transfer - a critical parameter for the performance of the main memory is its ability to read or write blocks of data at high speed.

Semiconductor RAM Memories

Internal organization of memory chips

- RAM – any location can be accessed for a Read or Write operation in some fixed amount of time that is independent of the location's address.
- Each memory cell can hold one bit of information.
- Memory cells are organized in the form of an array.
- One row is one memory word.
- All cells of a row are connected to a common line, known as the "word line".
- Word line is connected to the address decoder.
- Sense/write circuits are connected to the data input/output lines of the memory chip.

Internal Organization of Memory Chips



16 words of 8 bits each: 16x8 memory org.. It has 16 external connections: addr. 4, data 8, control: 2, power/ground: 2

1K memory cells: 128x8 memory, external connections: ? 19(7+8+2+2)

1Kx1:? 15 (10+1+2+2)

Figure 8.2. Organization of bit cells in a memory chip.

A Memory Chip

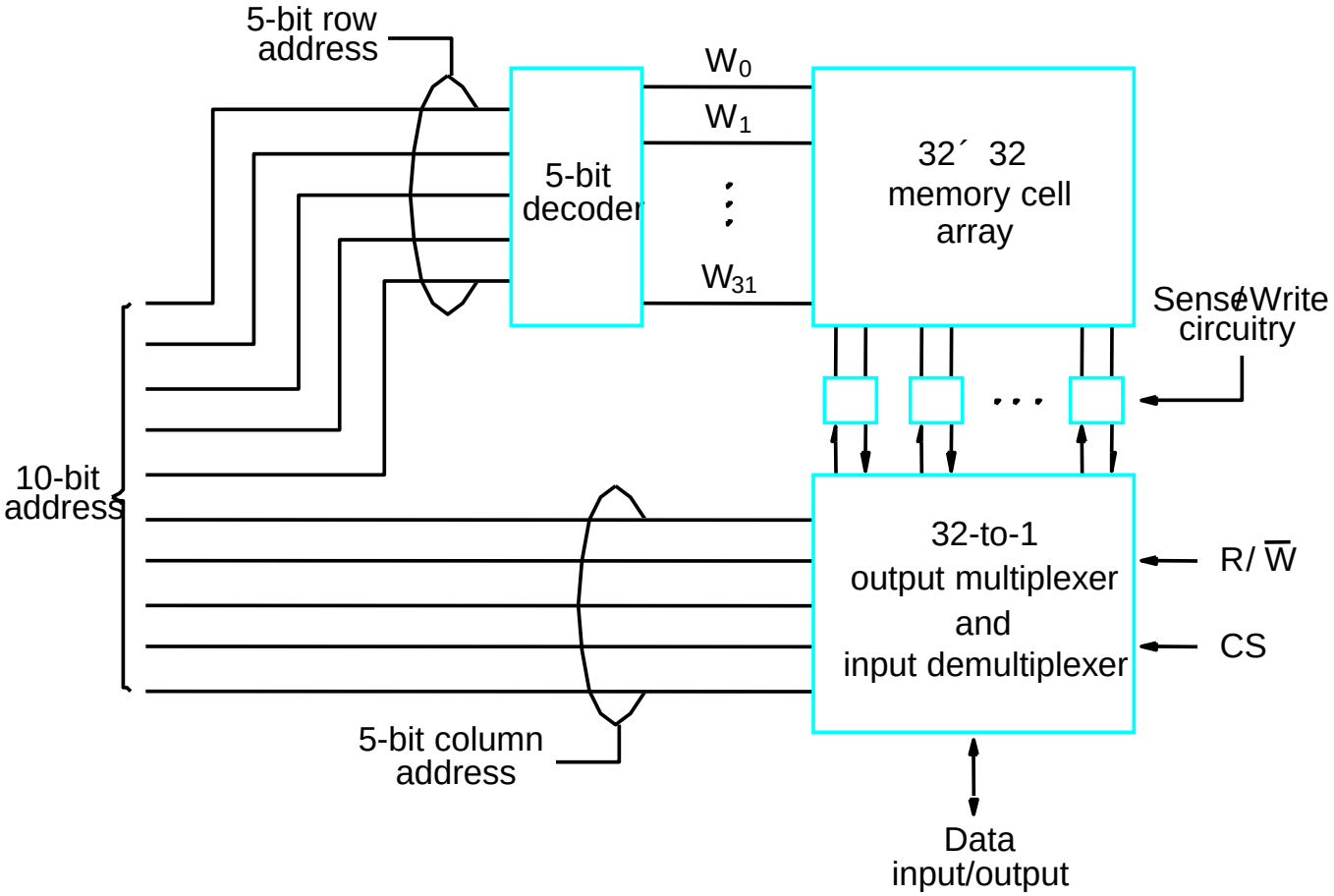


Figure 8.3. Organization of a 1K × 1 memory chip.

Static Memories

- The circuits are capable of retaining their state as long as power is applied.

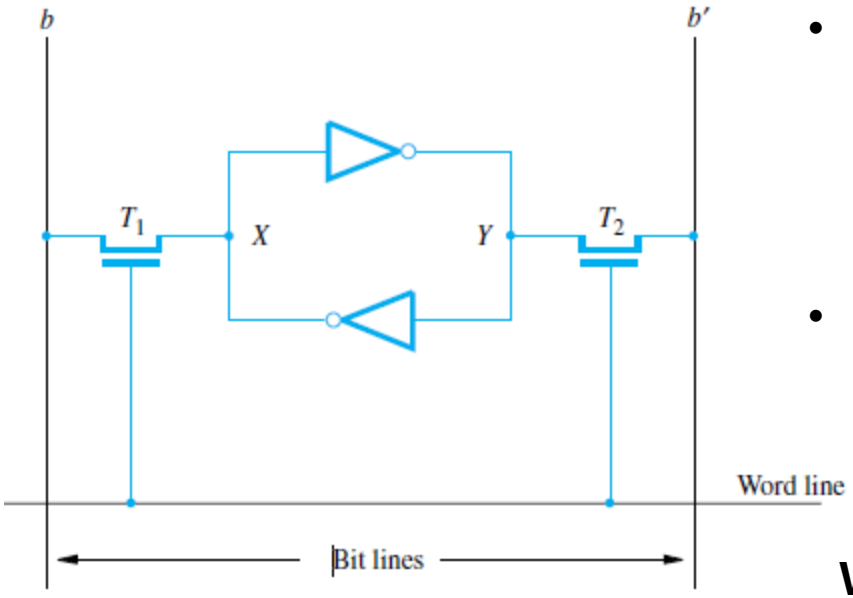


Figure 8.4 A static RAM cell.

Read Operation

- In order to read the state of the SRAM cell, the word line is activated to close switches *T*1 and *T*2. If the cell is in state 1, the signal on bit line *b* is high and the signal on bit line *b*′ is low.
- The Sense/Write circuit at the end of the two bit lines monitors their state and sets the corresponding output accordingly.

Write Operation

- During a Write operation, the Sense/Write circuit drives bit lines *b* and *b*′, instead of sensing their state.
- It places the appropriate value on bit line *b* and its complement on *b* and activates the word line.

Static Memories

- CMOS cell: low power consumption

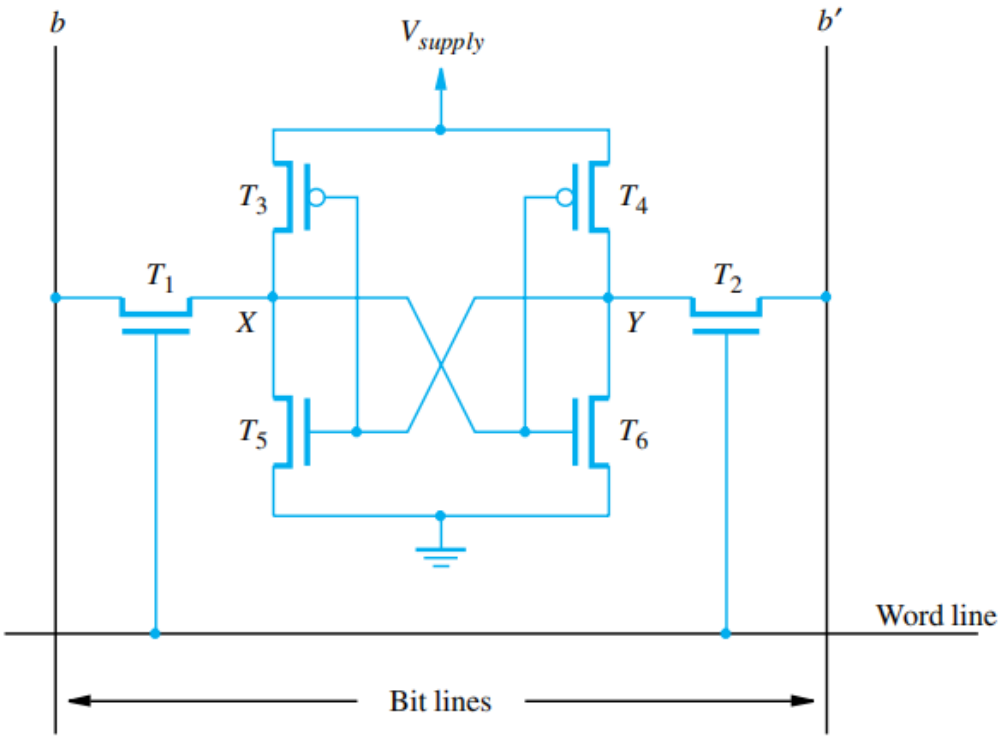


Figure 8.5 An example of a CMOS memory cell.

Dynamic RAMs

- Static RAMs are fast, but they cost more area and are more expensive.
- Dynamic RAMs (DRAMs) are cheap and area efficient, but they can not retain their state indefinitely – need to be periodically refreshed.

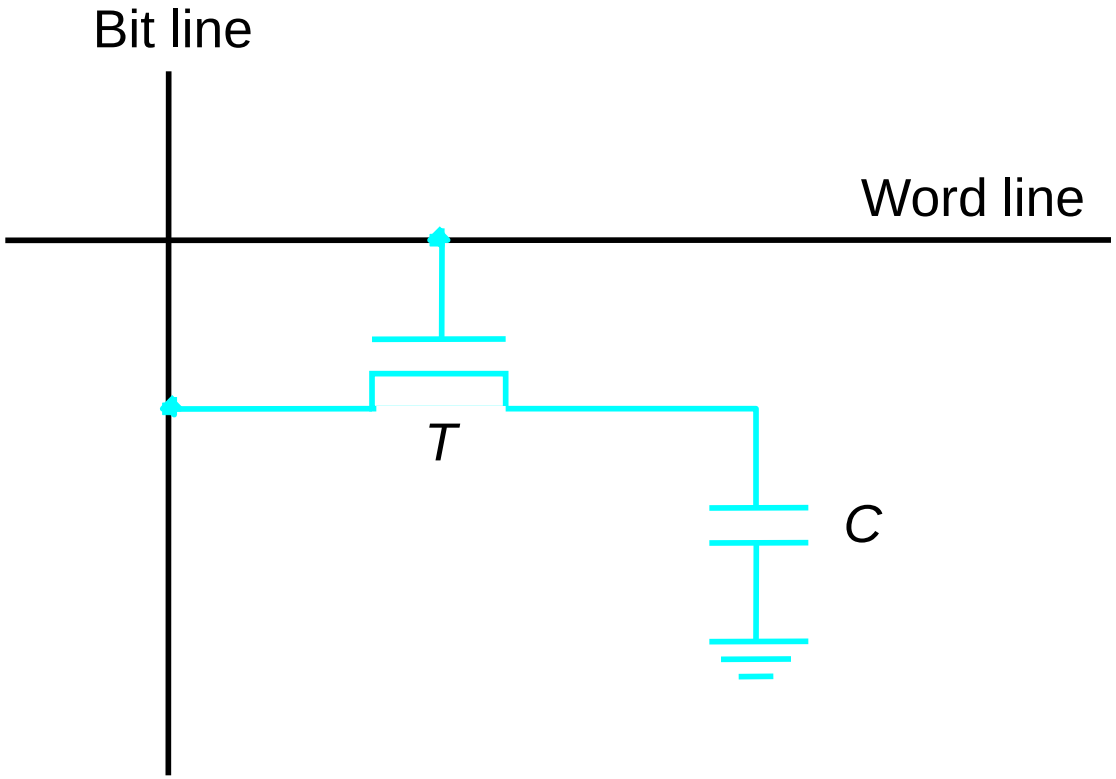


Figure 5.6. A single-transistor dynamic memory cell

A Dynamic Memory Chip

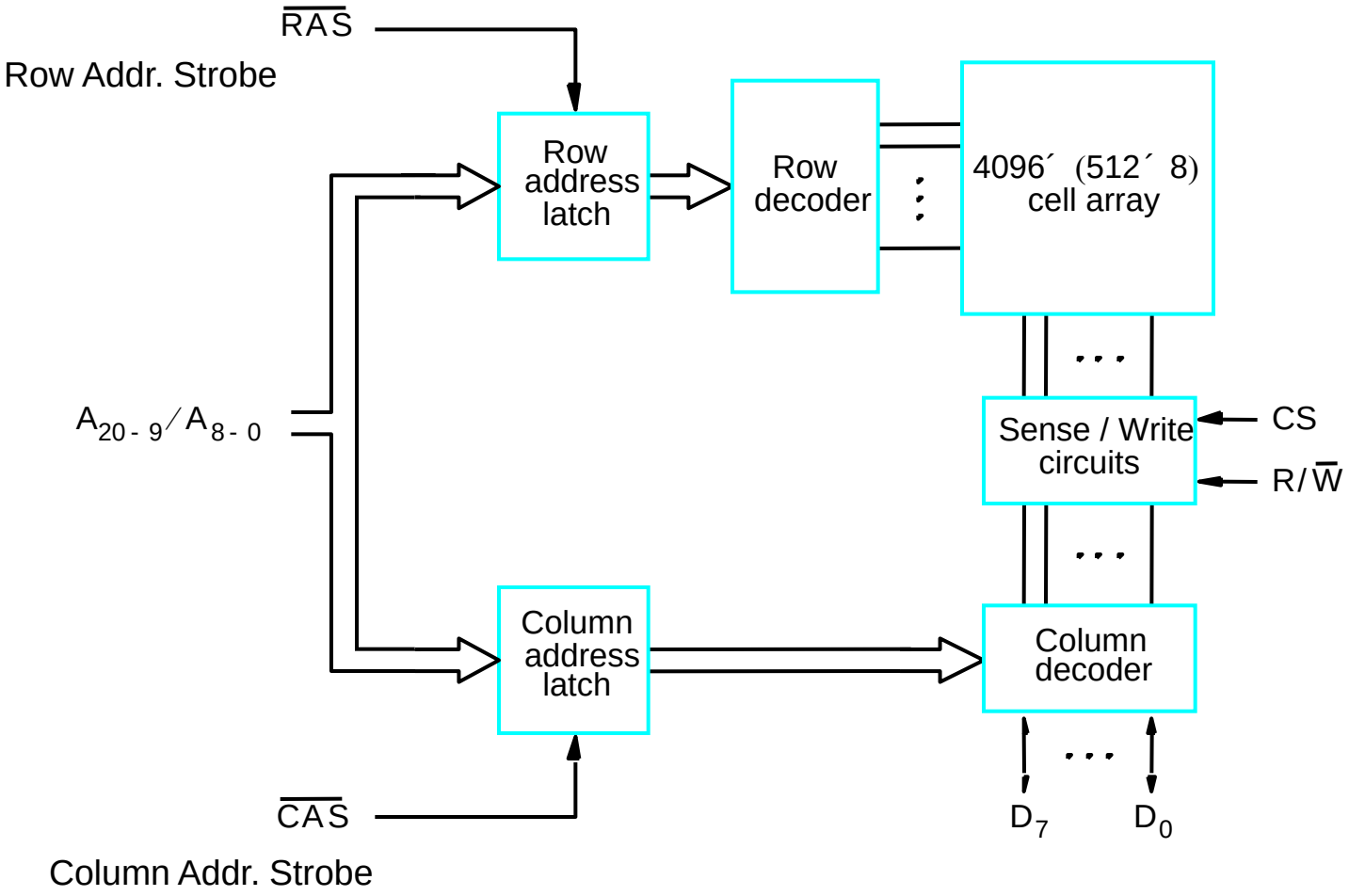


Figure 5.7. Internal organization of a 2M X 8 dynamic memory chip.

A Dynamic Memory Chip

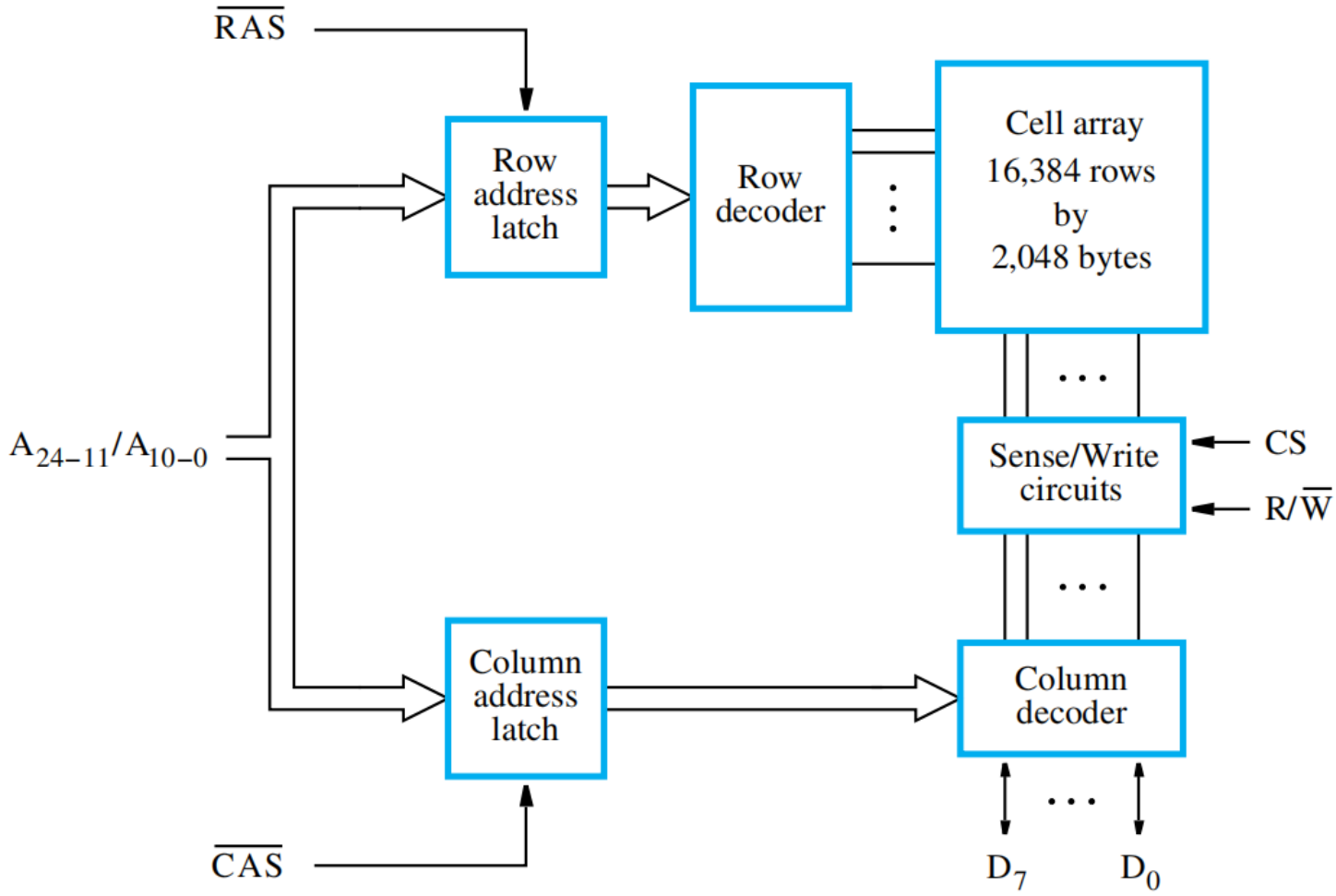


Figure 8.7 Internal organization of a 32M × 8 dynamic memory chip.

A Dynamic Memory Chip

- 14 address bits are neede to select a row, and another 11 bits are needed to specify a group of 8 bits in the selected row.
- To reduce the number of pins needed for external connections, the row and column addresses are multiplexed on 14 pins.
- During a Read or a Write operation, the row address is applied first and RAS signal is activated which strobes the address in latch
- Shortly after the row address is loaded, the column address is applied, stobed in latch with CAS control signal and the appropriate group of 8 Sense/Write circuits is selected for read or write operation depending on the signal in R/W.
- The timing of the operation is controlled by RAS and CAS generated by memory controller external to the chip - **Asynchronous DRAM**s.

Fast Page Mode

- When the DRAM in last slide is accessed, the contents of all 2048 bytes in the selected row are sensed, but only 8 bits are placed on the data lines D_{7-0}, as selected by A_{10-0}.
- A simple addition to the circuit makes it possible to access the other bytes in the same row without having to reselect the row.
- When a row address is applied, the contents of all cells in the selected row are loaded into the corresponding sense amplifier which also act as latches. Then, it is only necessary to apply different column addresses to place the different bytes on the data lines.
- Fast page mode – make it possible to access the other bytes in the same row without having to reselect the row.
- Good for bulk transfer at much faster rate.

Synchronous DRAMs

- The operations of SDRAM are controlled by a clock signal.

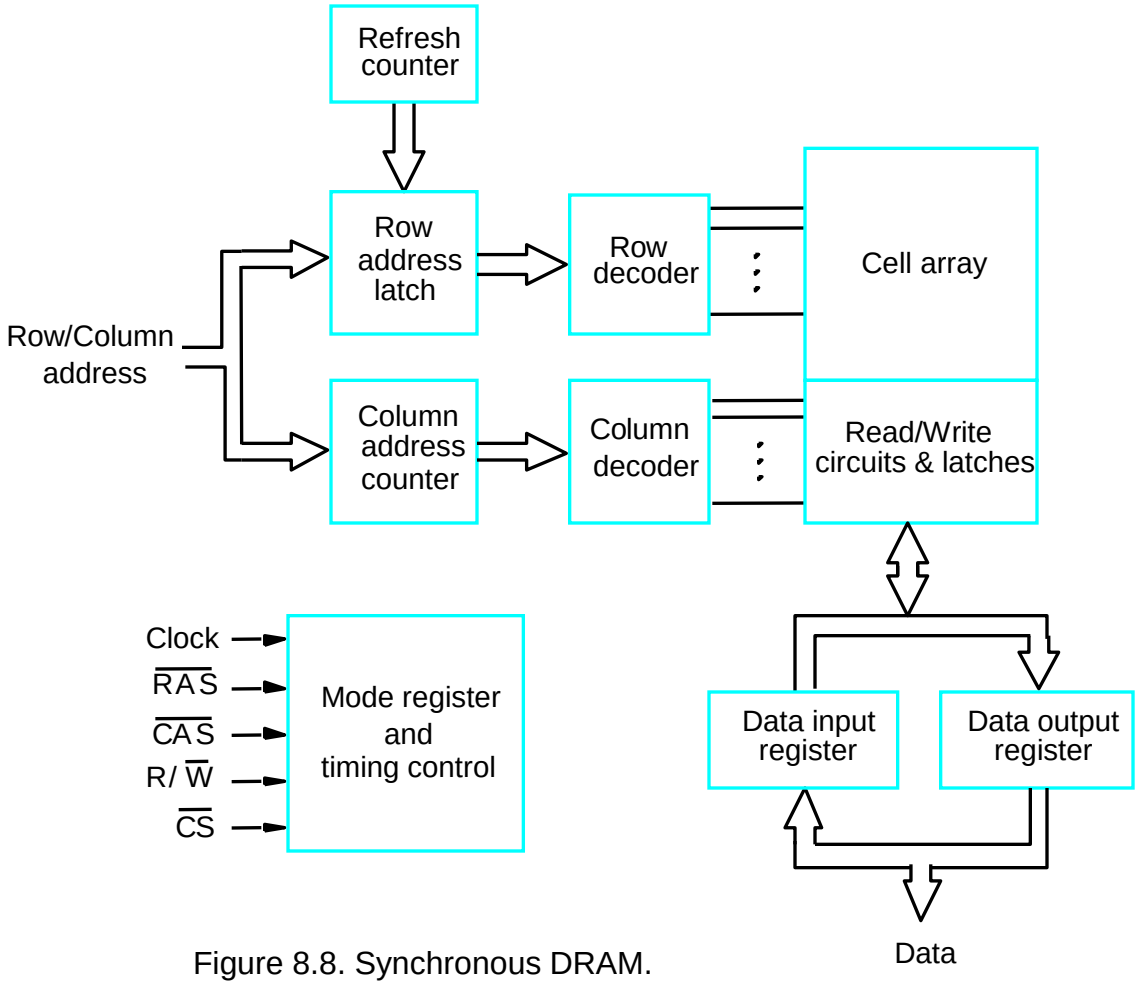


Figure 8.8. Synchronous DRAM.

Synchronous DRAMs

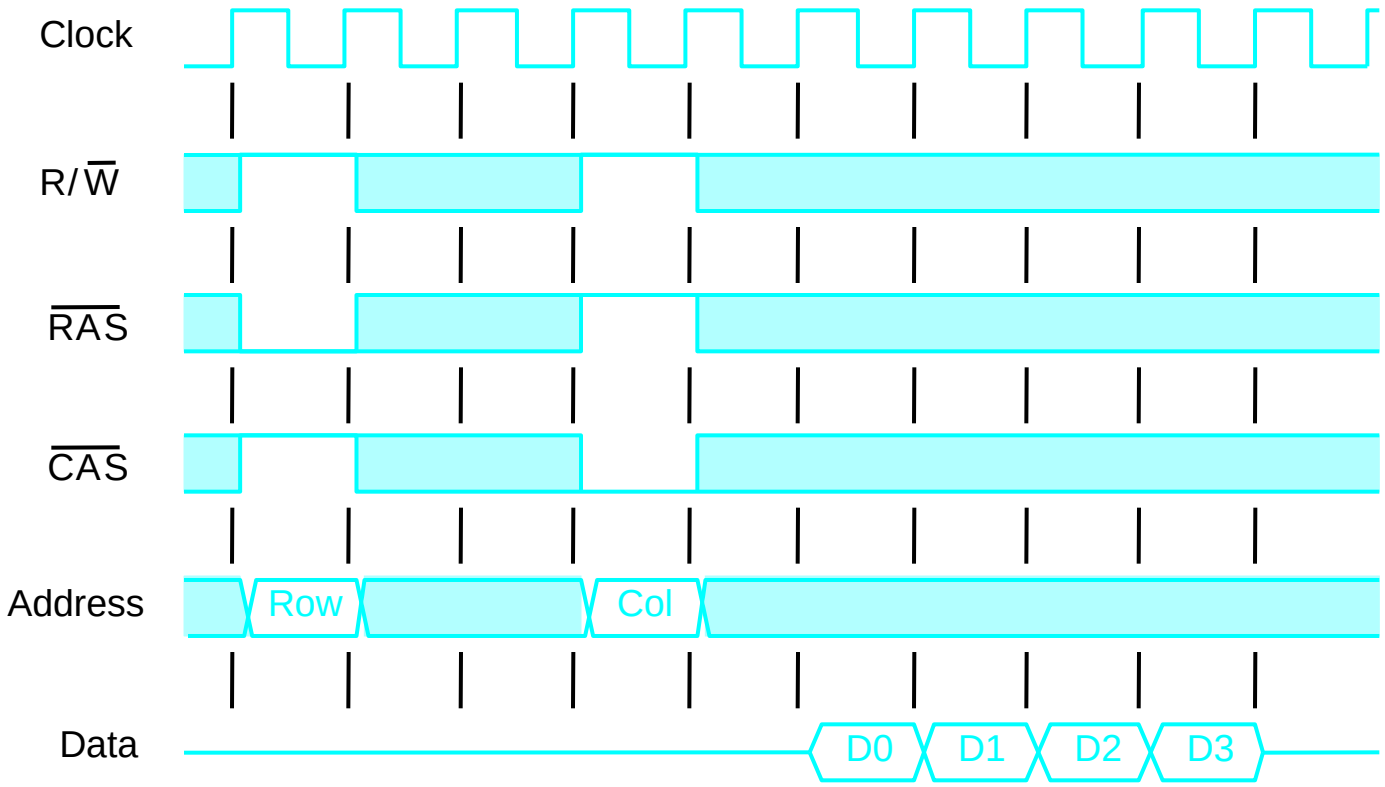


Figure 8.9. Burst read of length 4 in an SDRAM.

Synchronous DRAMs

- No CAS pulses is needed in burst operation.
- Refresh circuits are included.
- Synchronous DRAMs can deliver data at a very high rate, because all the control signals needed are generated inside the chip.
- The initial commercial SDRAMs in the 1990s were designed for clock speeds of up to 133 MHz.
- As technology evolved, much faster SDRAM chips were developed. Today’s SDRAMs operate with clock speeds that can exceed 1 GHz.

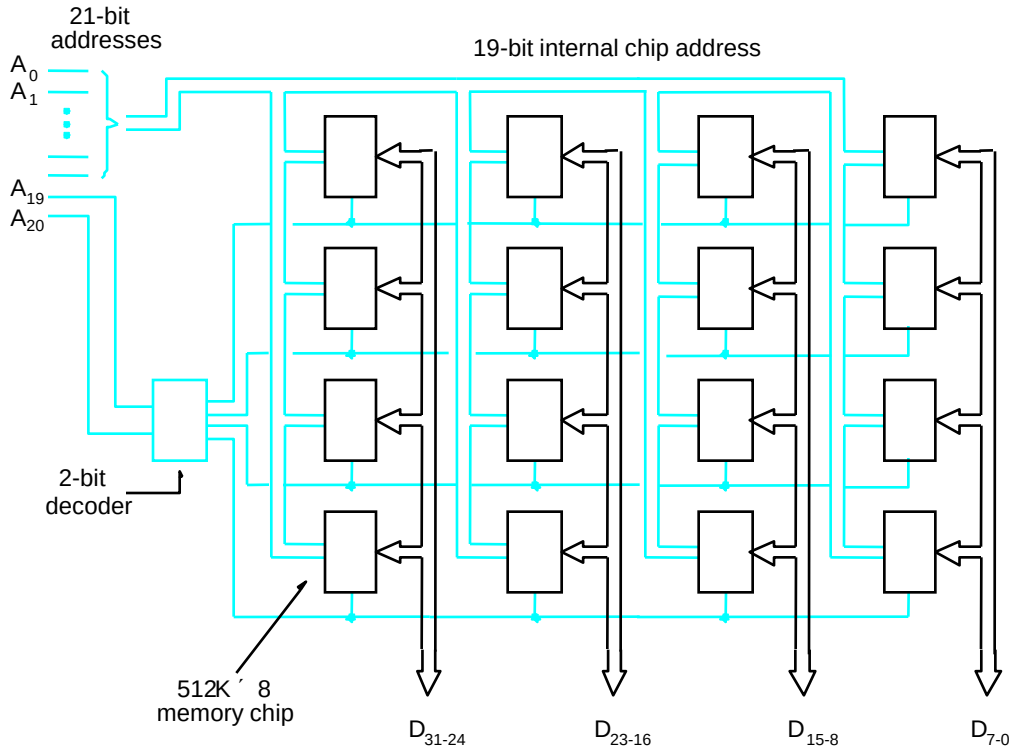
Latency and Bandwidth

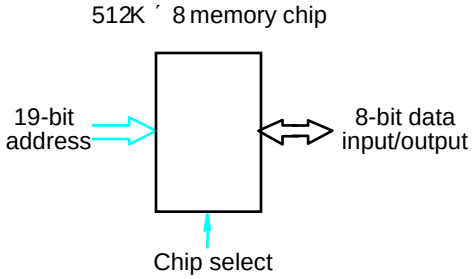
- The speed and efficiency of data transfers among memory, processor, and disk have a large impact on the performance of a computer system.
- Memory latency – the amount of time it takes to transfer a word of data to or from the memory.
- During block transfers, memory latency is the amount of time it takes to transfer the first word of a block.
- Memory bandwidth – the number of bits or bytes that can be transferred in one second. It is used to measure how much time is needed to transfer an entire block of data.
- Bandwidth is not determined solely by memory. It is the product of the rate at which data are transferred (and accessed) and the width of the data bus.

DDR SDRAM

- Double-Data-Rate SDRAM
- Standard SDRAM performs all actions on the rising edge of the clock signal.
- DDR SDRAM accesses the cell array in the same way, but transfers the data on both edges of the clock.
- The cell array is organized in two banks. Each can be accessed separately.
- DDR SDRAMs and standard SDRAMs are most efficiently used in applications where block transfers are prevalent.

Structures of Larger Memories





Memory System Considerations

- The choice of a RAM chip for a given application depends on several factors:

  Cost, speed, power, size…
- SRAMs are faster, more expensive, smaller.
- DRAMs are slower, cheaper, larger.
- Which one for cache and main memory, respectively?
- Refresh overhead – suppose a SDRAM whose cells are in 8K rows; 4 clock cycles are needed to access each row; then it takes $8192 \times 4 = 32{,}768$ cycles to refresh all rows; if the clock rate is 133 MHz, then it takes $32{,}768/(133 \times 10^{-6}) = 246 \times 10^{-6}$ seconds; suppose the typical refreshing period is 64 ms, then the refresh overhead is $0.246/64 = 0.0038 < 0.4\%$ of the total time available for accessing the memory.

Memory Controller

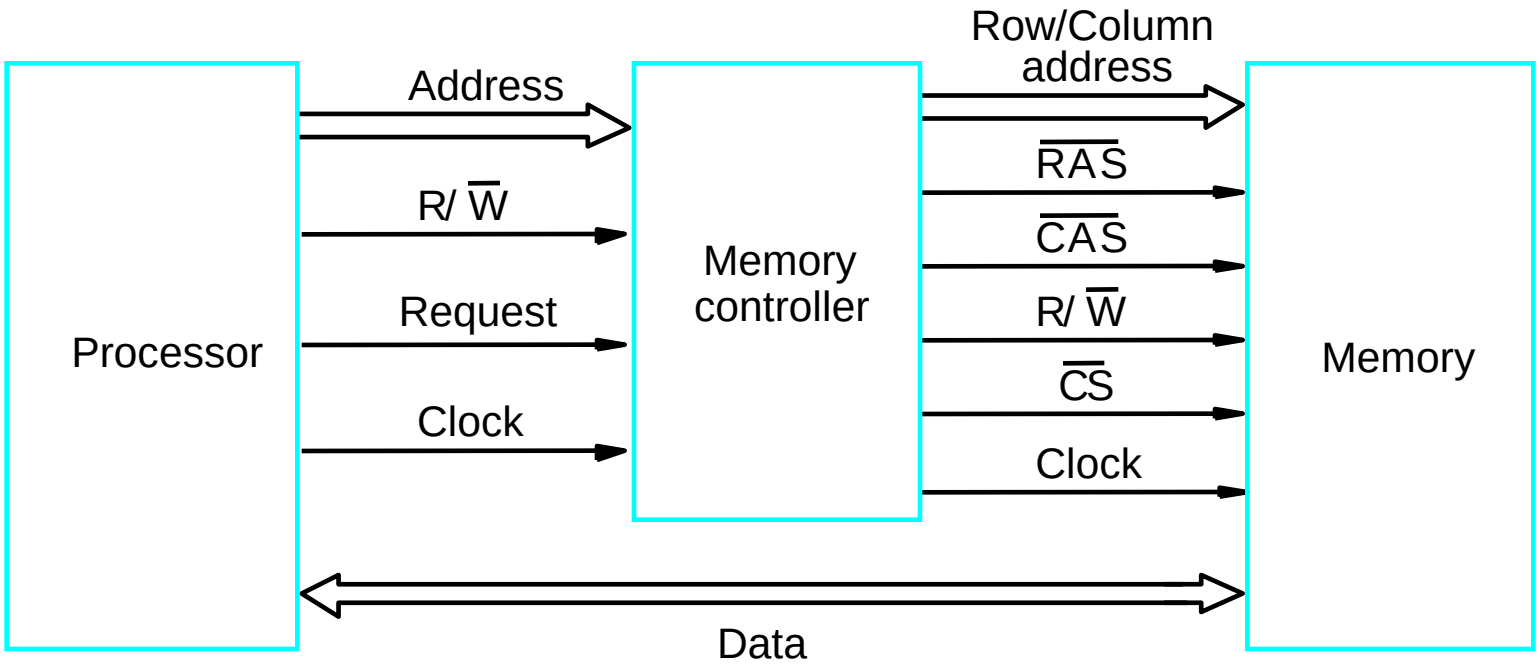


Figure 5.11. Use of a memory controller.

Memory controller

- Recall that in a dynamic memory chip, to reduce the number of pins, multiplexed addresses are used.
- Address is divided into two parts:
  - High-order address bits select a row in the array.
  - They are provided first, and latched using RAS signal.
  - Low-order address bits select a column in the row.
  - They are provided later, and latched using CAS signal.
- However, a processor issues all address bits at the same time.
- In order to achieve the multiplexing, memory controller circuit is inserted between the processor and memory.

Read-Only Memories

Read-Only Memories (ROMs)

- SRAM and SDRAM chips are volatile:
  - Lose the contents when the power is turned off.
- Many applications need memory devices to retain contents after the power is turned off.
  - For example, computer is turned on, the operating system must be loaded from the disk into the memory.
  - Store instructions which would load the OS from the disk.
  - Need to store these instructions so that they will not be lost after the power is turned off.
  - We need to store the instructions into a non-volatile memory.
- Non-volatile memory is read in the same manner as volatile memory.
  - Separate writing process is needed to place information in this memory.
  - Normal operation involves only reading of data, this type of memory is called Read-Only memory (ROM).

Read-Only-Memory

- Volatile / non-volatile memory
- ROM
- PROM: programmable ROM
- EPROM: erasable, reprogrammable ROM
- EEPROM: can be programmed and erased electrically

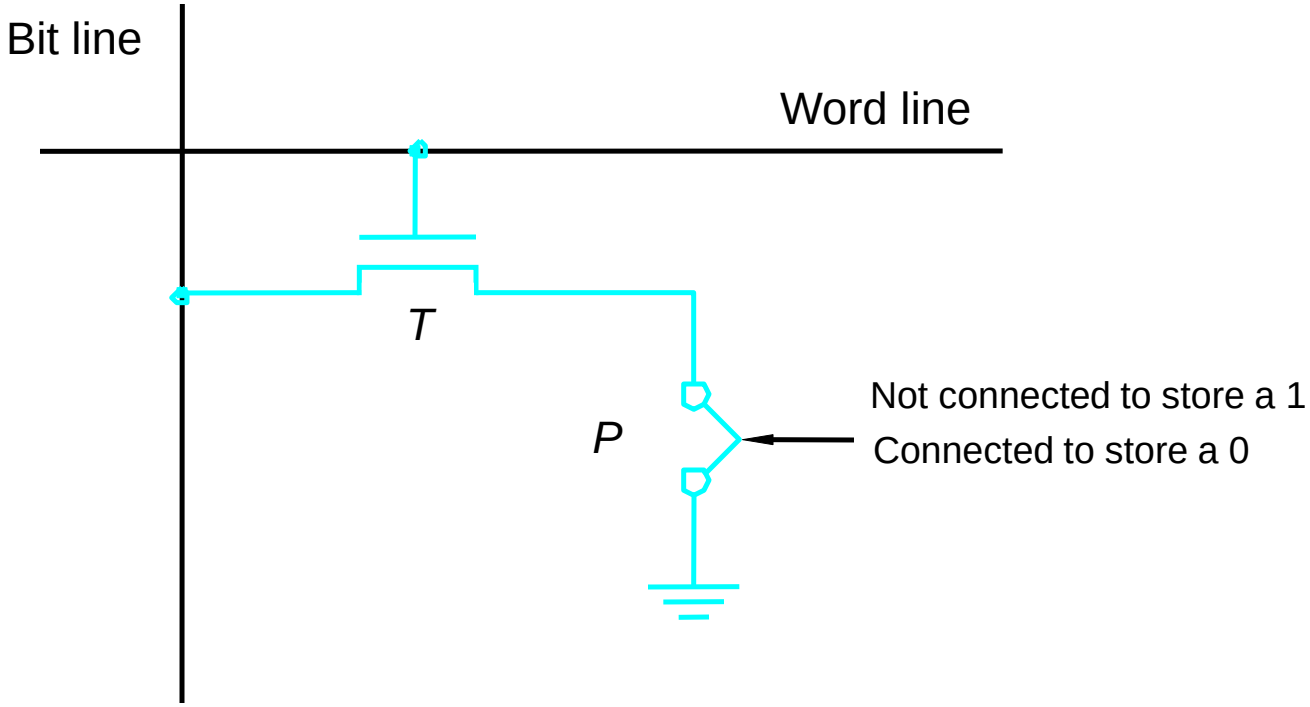


Figure 5.12. A ROM cell.

Read-Only Memories

- Read-Only Memory:
  - Data are written into a ROM when it is manufactured.
- Programmable Read-Only Memory (PROM):
  - Allow the data to be loaded by a user.
  - Process of inserting the data is irreversible.
  - Storing information specific to a user in a ROM is expensive.
  - Providing programming capability to a user may be better.
- Erasable Programmable Read-Only Memory (EPROM):
  - Stored data to be erased and new data to be loaded.
  - Flexibility, useful during the development phase of digital systems.
  - Erasable, reprogrammable ROM.
  - Erasure requires exposing the ROM to UV light.

Read-Only Memories

- Electrically Erasable Programmable Read-Only Memory (EEPROM):
  - To erase the contents of EPROMs, they have to be exposed to ultraviolet light.
  - Physically removed from the circuit.
  - EEPROMs the contents can be stored and erased electrically.
- Flash memory:
  - Has similar approach to EEPROM.
  - Read the contents of a single cell, but write the contents of an entire block of cells.
  - Flash devices have greater density. Higher capacity and low storage cost per bit.
  - Power consumption of flash memory is very low, making it attractive for use in equipment that is battery-driven.
  - Single flash chips are not sufficiently large, so larger memory modules are implemented using flash cards and flash drives.

Direct Memory Access

Direct Memory Access (contd..)

- Direct Memory Access (DMA):
  - A special control unit may be provided to transfer a block of data directly between an I/O device and the main memory, without continuous intervention by the processor.
- Control unit which performs these transfers is a part of the I/O device's interface circuit. This control unit is called as a DMA controller.
- DMA controller performs functions that would be normally carried out by the processor:
  - For each word, it provides the memory address and all the control signals.
  - To transfer a block of data, it increments the memory addresses and keeps track of the number of transfers.

Direct Memory Access (contd..)

- DMA controller can transfer a block of data from an external device to the processor, without any intervention from the processor.
  - However, the operation of the DMA controller must be under the control of a program executed by the processor. That is, the processor must initiate the DMA transfer.
- To initiate the DMA transfer, the processor informs the DMA controller of:
  - Starting address,
  - Number of words in the block.
  - Direction of transfer (I/O device to the memory, or memory to the I/O device).
- Once the DMA controller completes the DMA transfer, it informs the processor by raising an interrupt signal.

Direct Memory Access (contd..)

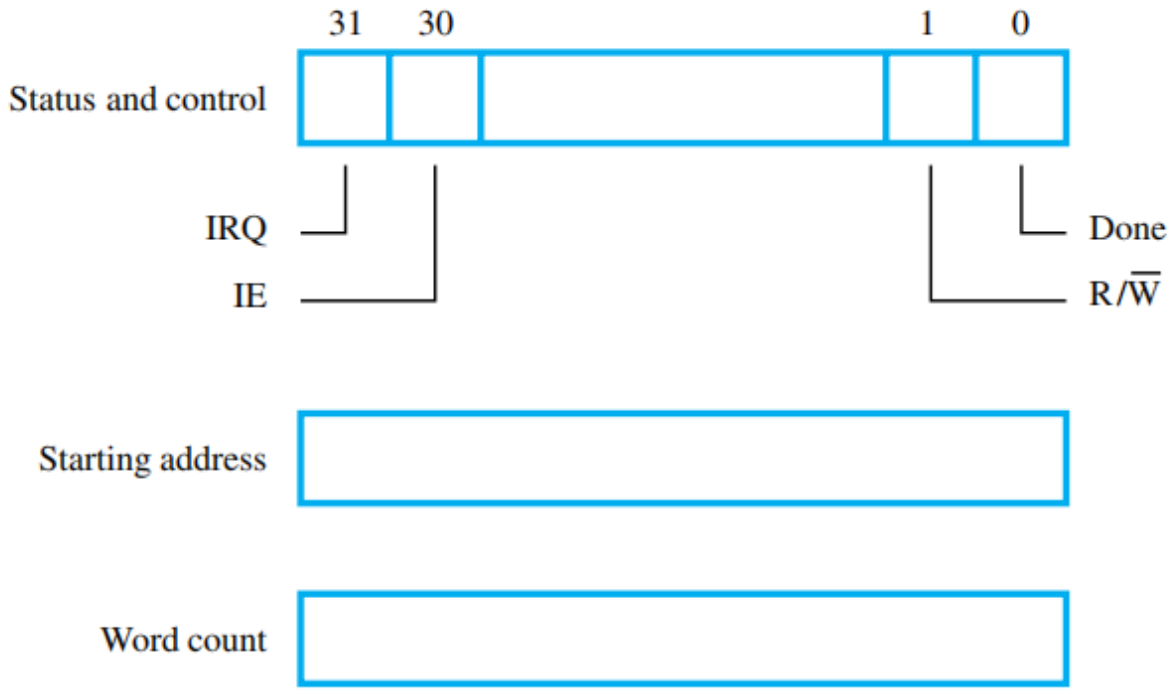


Figure 8.12 Typical registers in a DMA controller.

Direct Memory Access

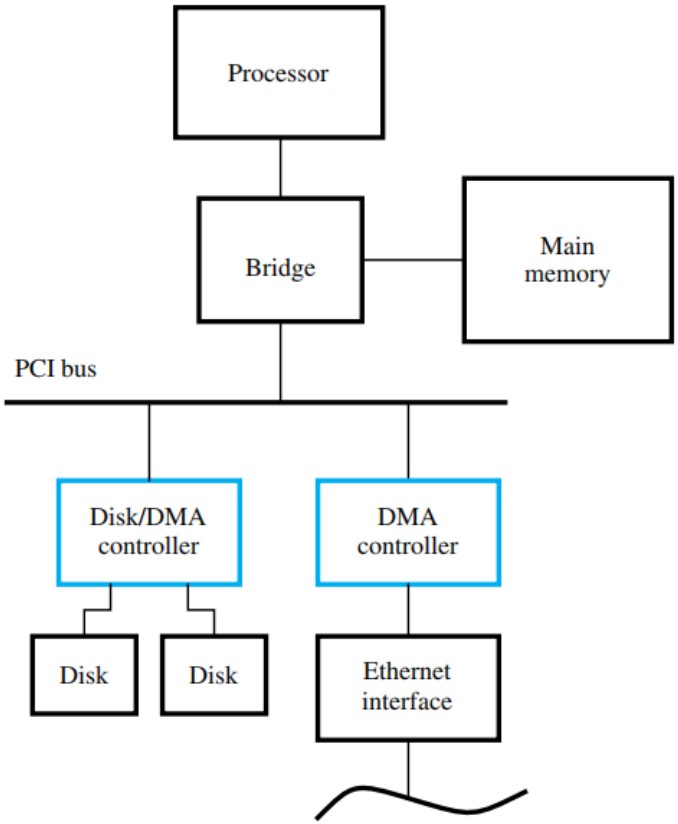


Figure 8.13 Use of DMA controllers in a computer system.

- *DMA controller connects a high-speed network to the computer bus.*
- *Disk controller, which controls two disks also has DMA capability. It provides two DMA channels.*
- *It can perform two independent DMA operations, as if each disk has its own DMA controller. The registers to store the memory address, word count and status and control information are duplicated.*

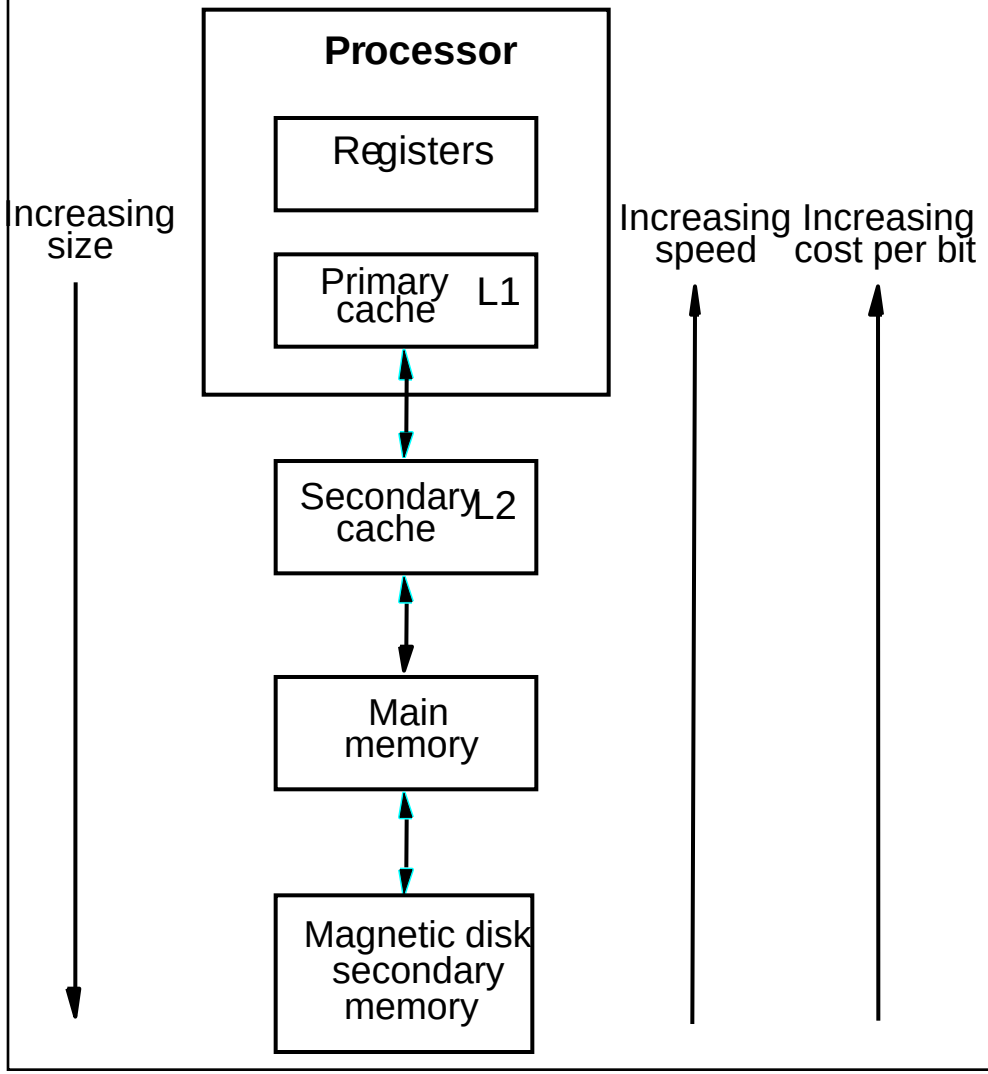
Direct Memory Access (contd..)

- Processor and DMA controllers have to use the bus in an interwoven fashion to access the memory.
  - DMA devices are given higher priority than the processor to access the bus.
  - Among different DMA devices, high priority is given to high-speed peripherals such as a disk or a graphics display device.
- Processor originates most memory access cycles on the bus.
  - DMA controller can be said to "steal" memory access cycles from the bus. This interweaving technique is called as "cycle stealing".
- An alternate approach is the provide a DMA controller an exclusive capability to initiate transfers on the bus, and hence exclusive access to the main memory. This is known as the block or burst mode.

Speed, Size, and Cost

- A big challenge in the design of a computer system is to provide a sufficiently large memory, with a reasonable speed at an affordable cost.
- Static RAM:
  - Very fast, but expensive, because a basic SRAM cell has a complex circuit making it impossible to pack a large number of cells onto a single chip.
- Dynamic RAM:
  - Simpler basic cell circuit, hence are much less expensive, but significantly slower than SRAMs.
- Magnetic disks:
  - Storage provided by DRAMs is higher than SRAMs, but is still less than what is necessary.
  - Secondary storage such as magnetic disks provide a large amount of storage, but is much slower than DRAMs.

Memory Hierarchy



- *Fastest access is to the data held in processor registers. Registers are at the top of the memory hierarchy.*
- *Relatively small amount of memory that can be implemented on the processor chip. This is processor cache.*
- *Two levels of cache. Level 1 (L1) cache is on the processor chip. Level 2 (L2) cache is in between main memory and processor.*
- *Next level is main memory, implemented as SIMMs. Much larger, but much slower than cache memory.*
- *Next level is magnetic disks. Huge amount of inexepensive storage.*
- *Speed of memory access is critical, the idea is to bring instructions and data that will be used in the near future as close to the processor as possible.*

Cache Memories

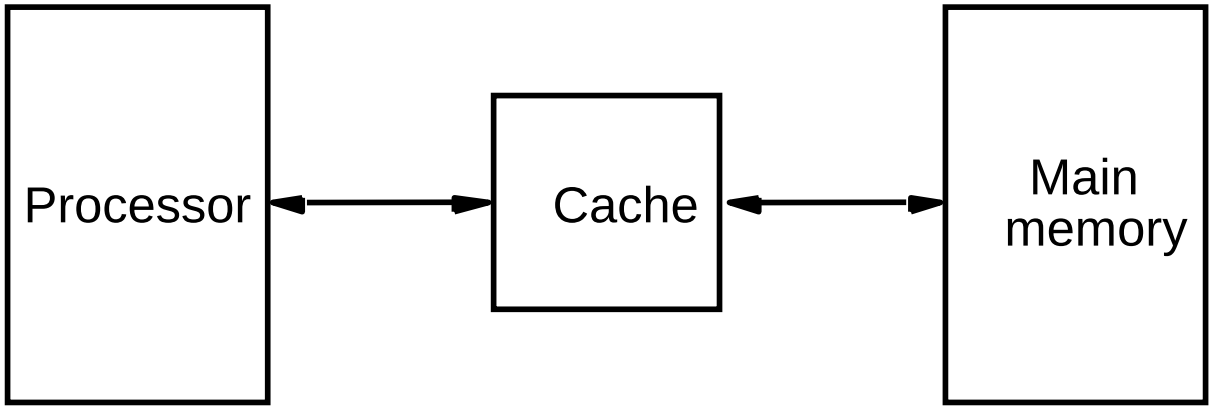
Cache Memories

- Processor is much faster than the main memory.
  - As a result, the processor has to spend much of its time waiting while instructions and data are being fetched from the main memory.
  - Major obstacle towards achieving good performance.
- Speed of the main memory cannot be increased beyond a certain point.
- Cache memory is an architectural arrangement which makes the main memory appear faster to the processor than it really is.
- Cache memory is based on the property of computer programs known as “locality of reference”.

Locality of Reference

- Analysis of programs indicates that many instructions in localized areas of a program are executed repeatedly during some period of time, while the others are accessed relatively less frequently.
  - These instructions may be the ones in a loop, nested loop or few procedures calling each other repeatedly.
  - This is called "locality of reference".
- Temporal locality of reference:
  - Recently executed instruction is likely to be executed again very soon.
- Spatial locality of reference:
  - Instructions with addresses close to a recently instruction are likely to be executed soon.

Cache memories



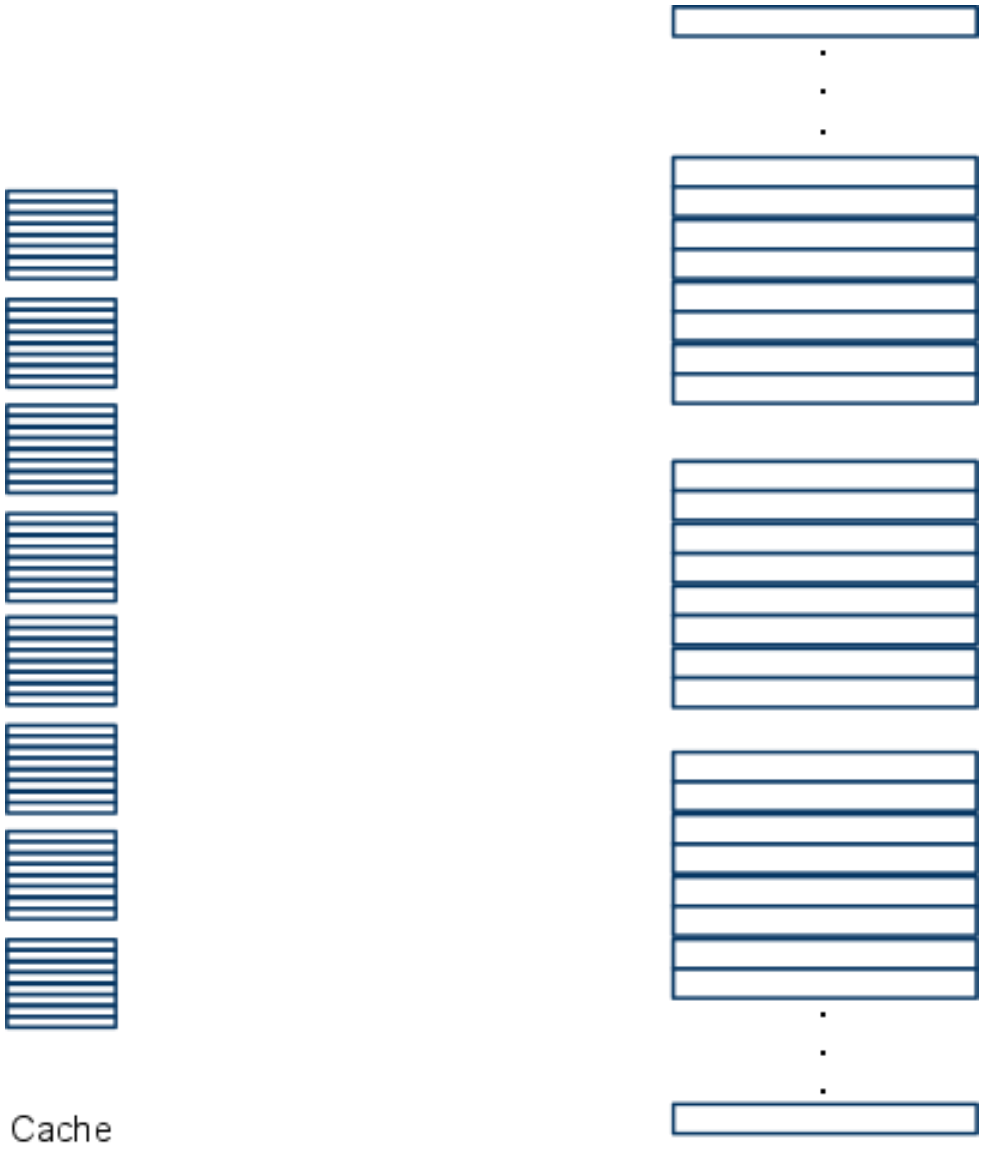
- *Processor issues a Read request, a block of words is transferred from the main memory to the cache, one word at a time.*
- *Subsequent references to the data in this block of words are found in the cache.*
- *At any given time, only some blocks in the main memory are held in the cache. Which blocks in the main memory are in the cache is determined by a "mapping function".*
- *When the cache is full, and a block of words needs to be transferred from the main memory, some block of words in the cache must be replaced. This is determined by a "replacement algorithm".*

Cache hit

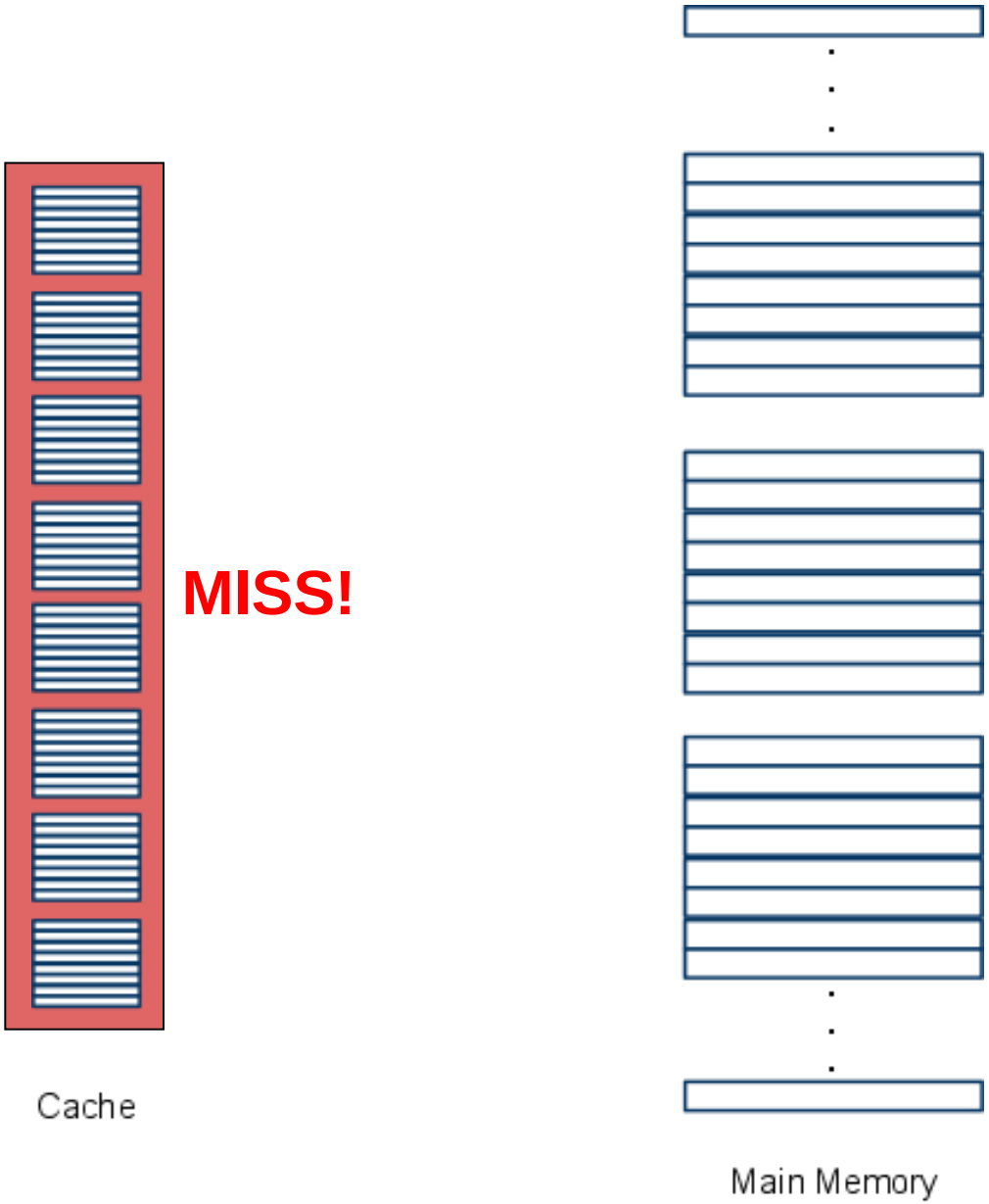
- *Existence of a cache is transparent to the processor. The processor issues Read and Write requests in the same manner.*
- *If the data is in the cache it is called a Read or Write hit.*
- *Read hit:*
  - *The data is obtained from the cache.*
- *Write hit:*
  - *Cache has a replica of the contents of the main memory.*
  - *Contents of the cache and the main memory may be updated simultaneously. This is the write-through protocol.*
  - *Update the contents of the cache, and mark it as updated by setting a bit known as the dirty bit or modified bit. The contents of the main memory are updated when this block is replaced. This is write-back or copy-back protocol.*

Cache miss

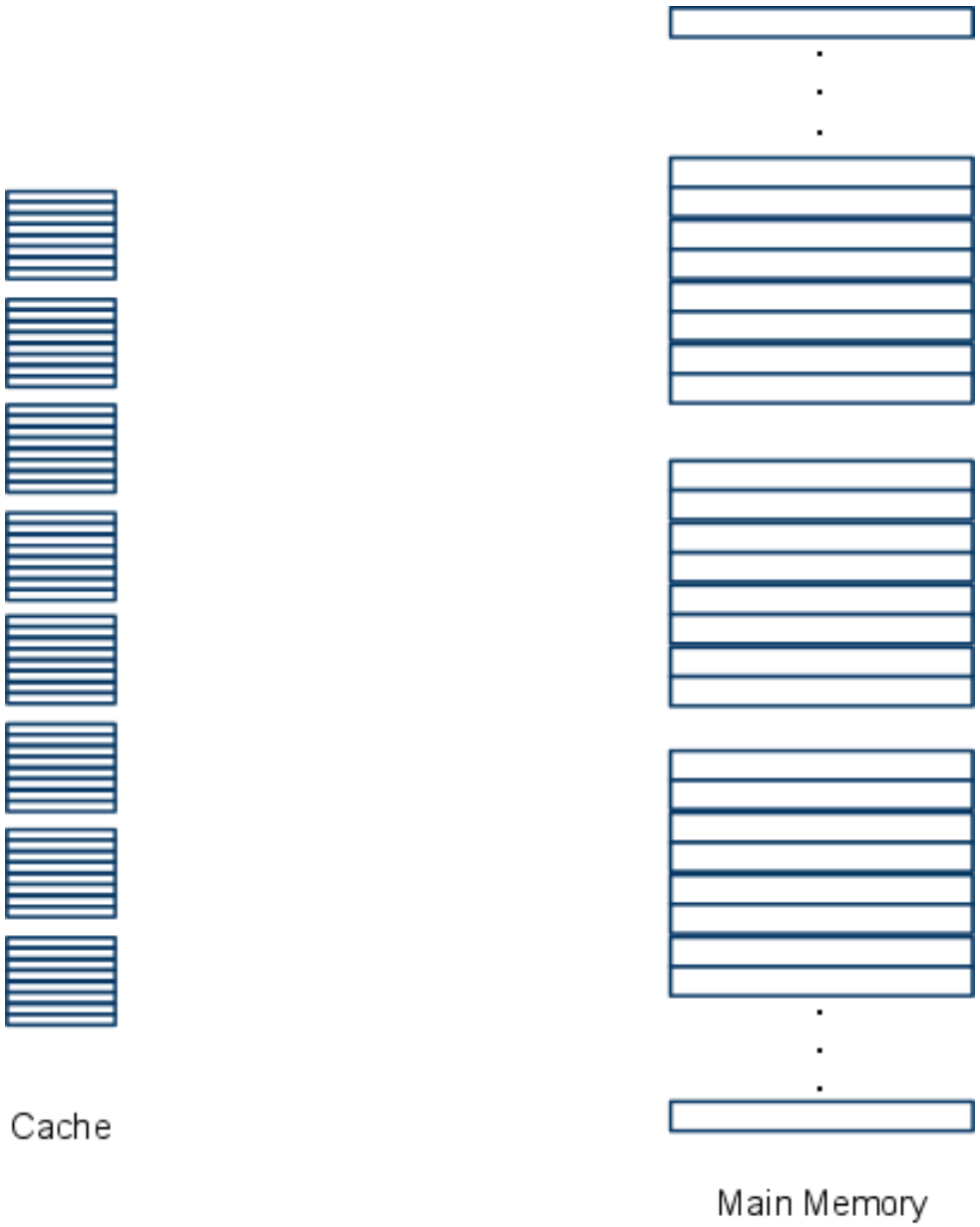
- *If the data is not present in the cache, then a Read miss or Write miss occurs.*
- *Read miss:*
  - *Block of words containing this requested word is transferred from the memory.*
  - *After the block is transferred, the desired word is forwarded to the processor.*
  - *The desired word may also be forwarded to the processor as soon as it is transferred without waiting for the entire block to be transferred. This is called load-through or early-restart.*
- *Write-miss:*
  - *Write-through protocol is used, then the contents of the main memory are updated directly.*
  - *If write-back protocol is used, the block containing the addressed word is first brought into the cache. The desired word is overwritten with new information.*



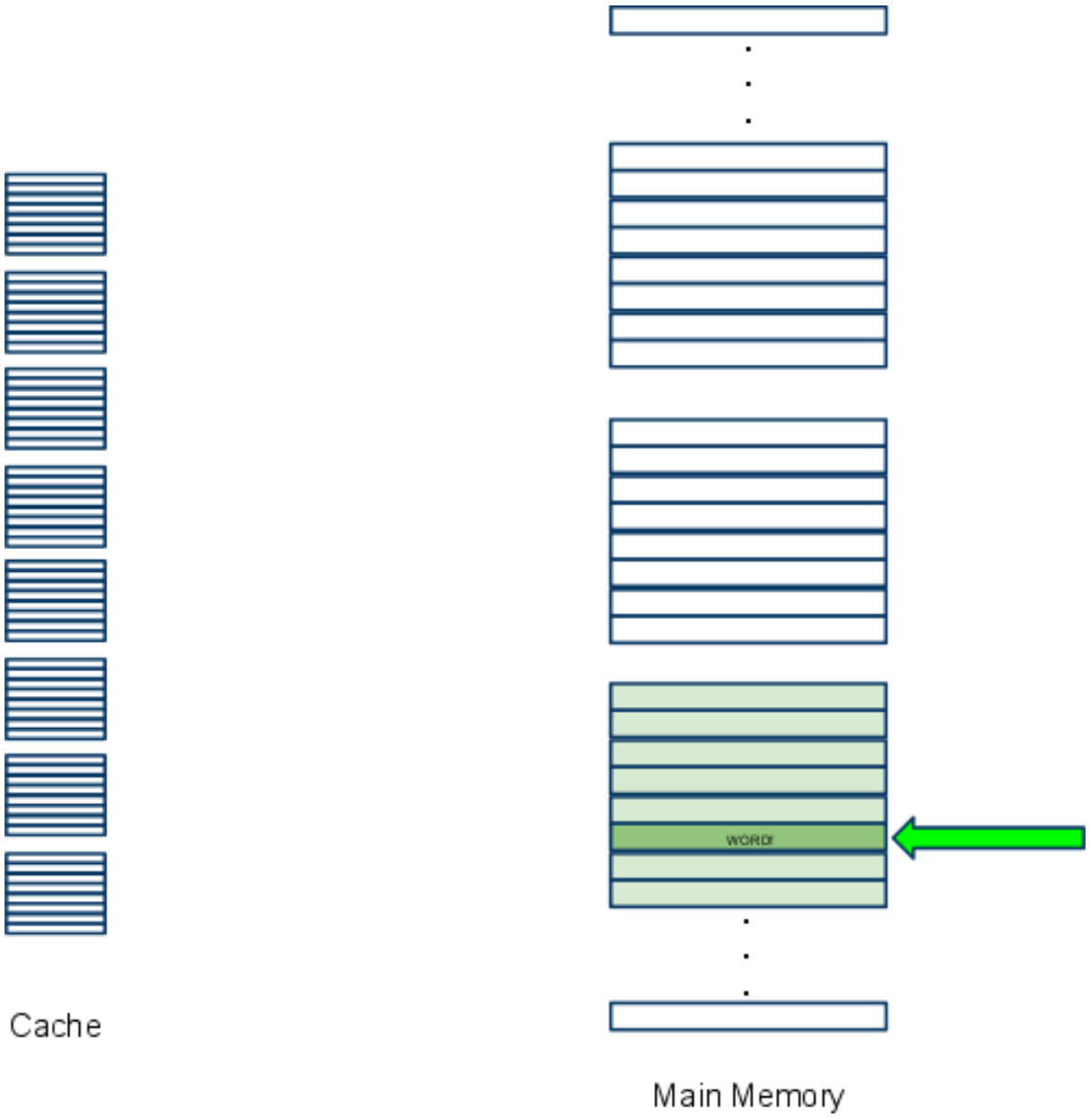
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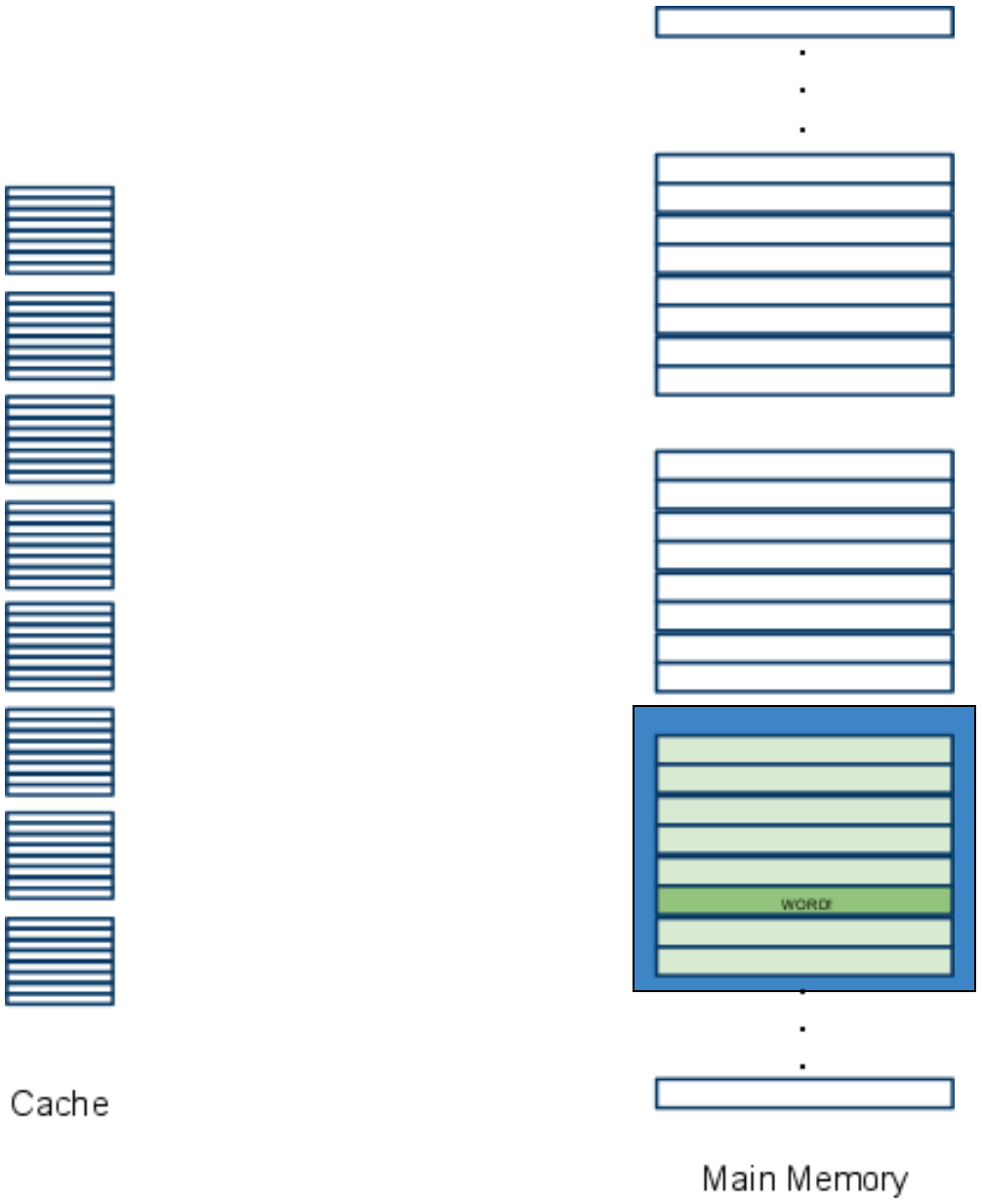
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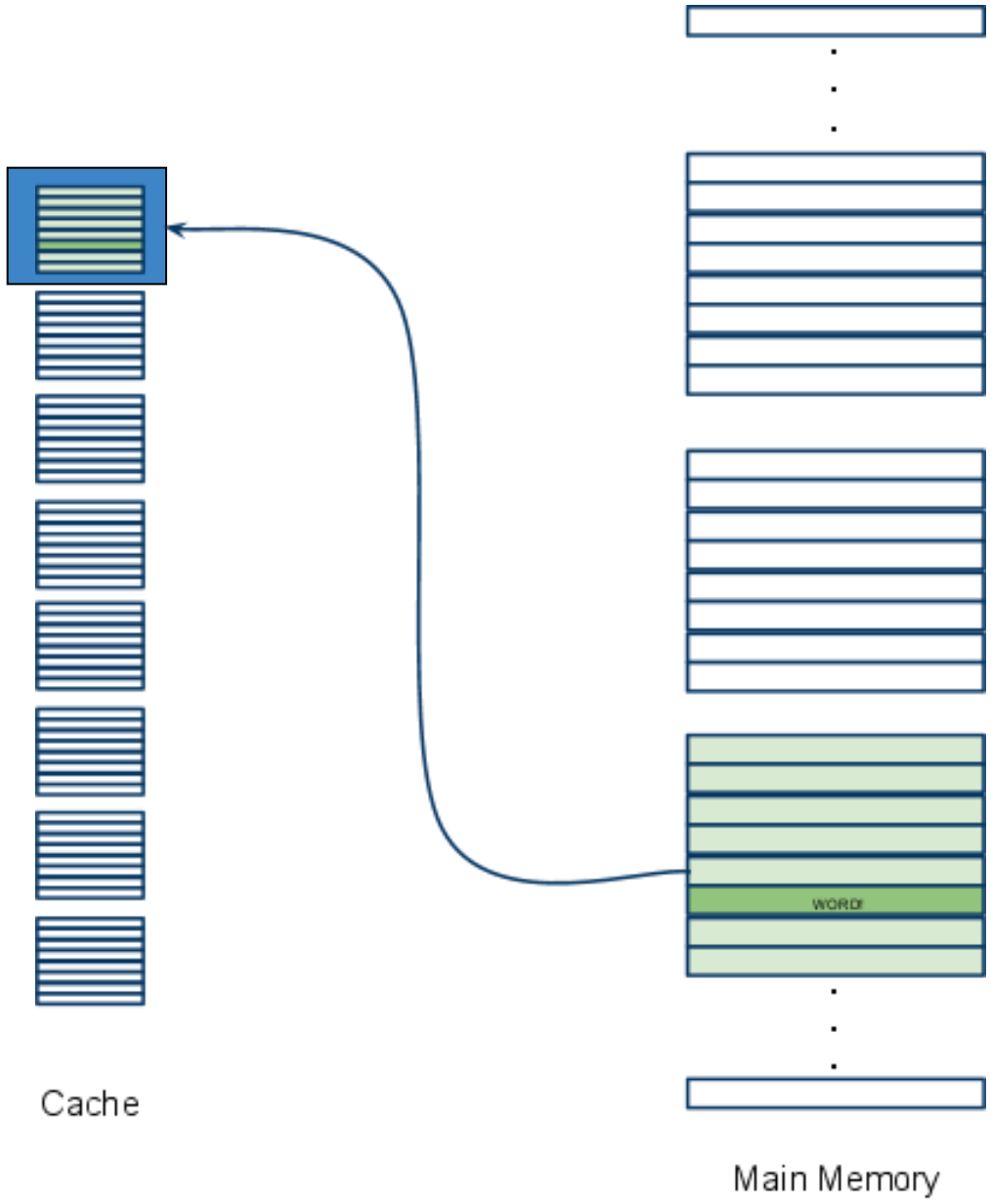
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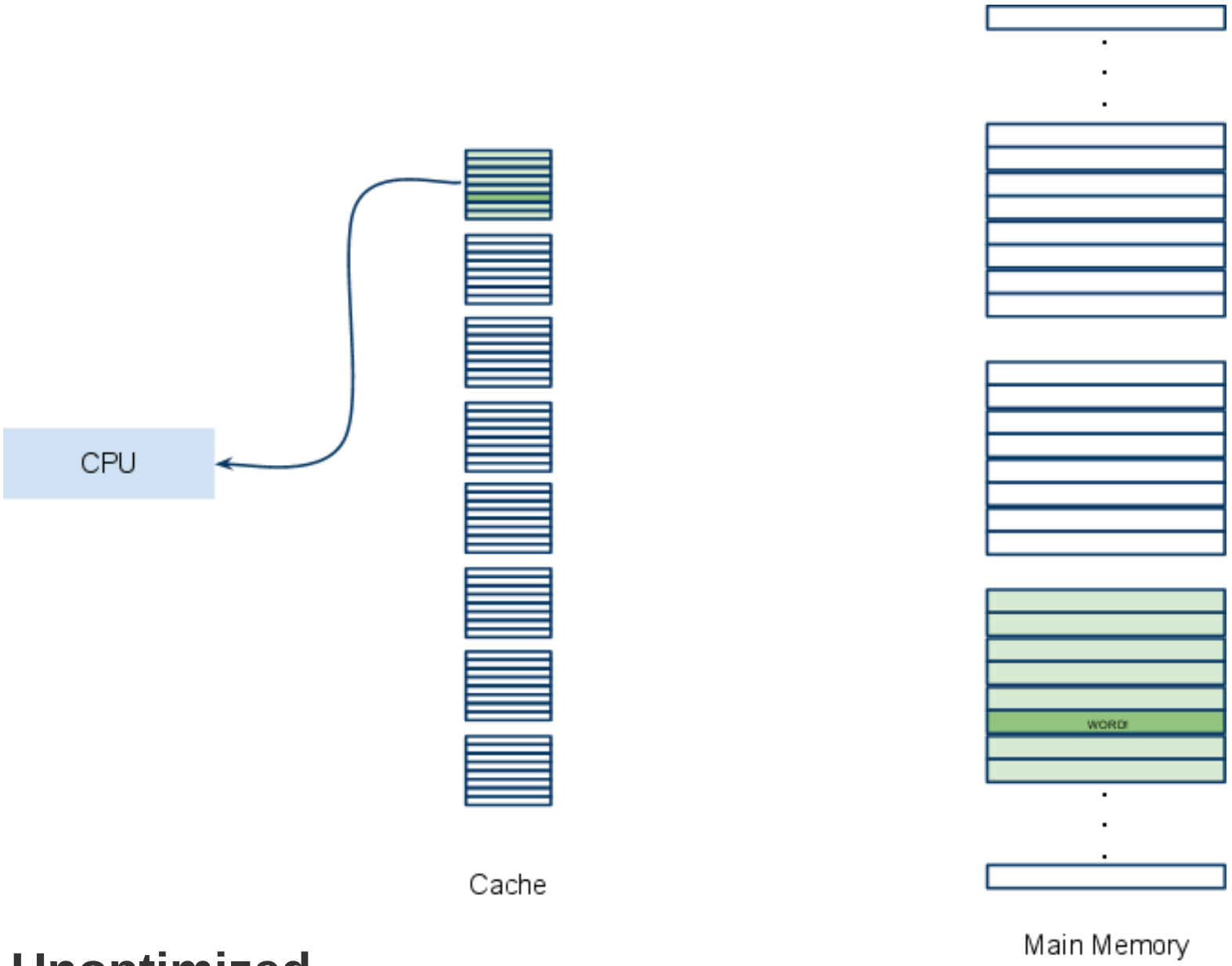
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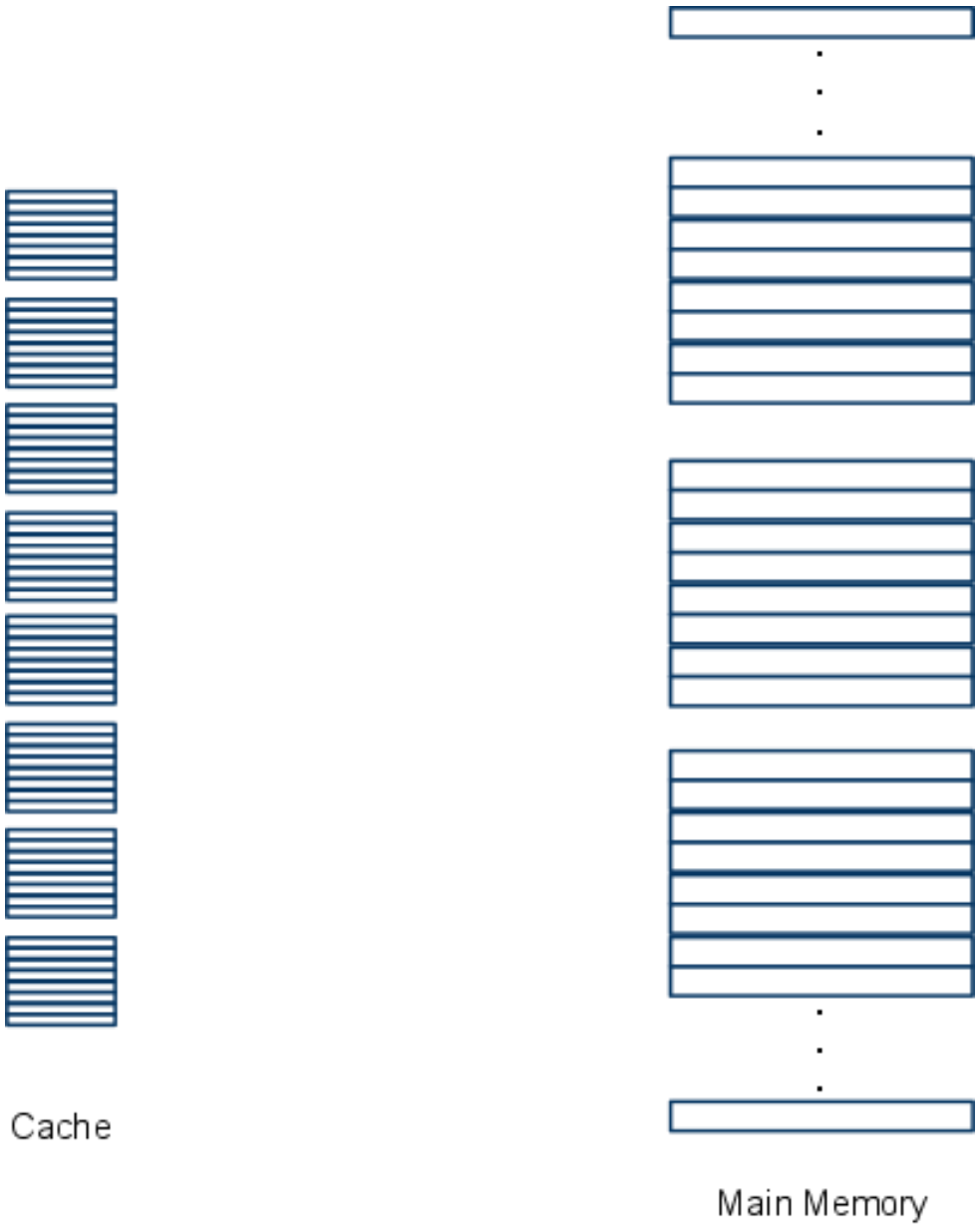
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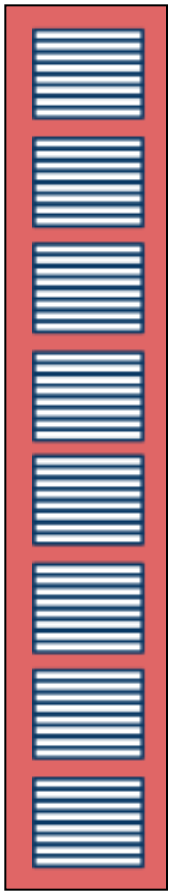
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Unoptimized



Early Start



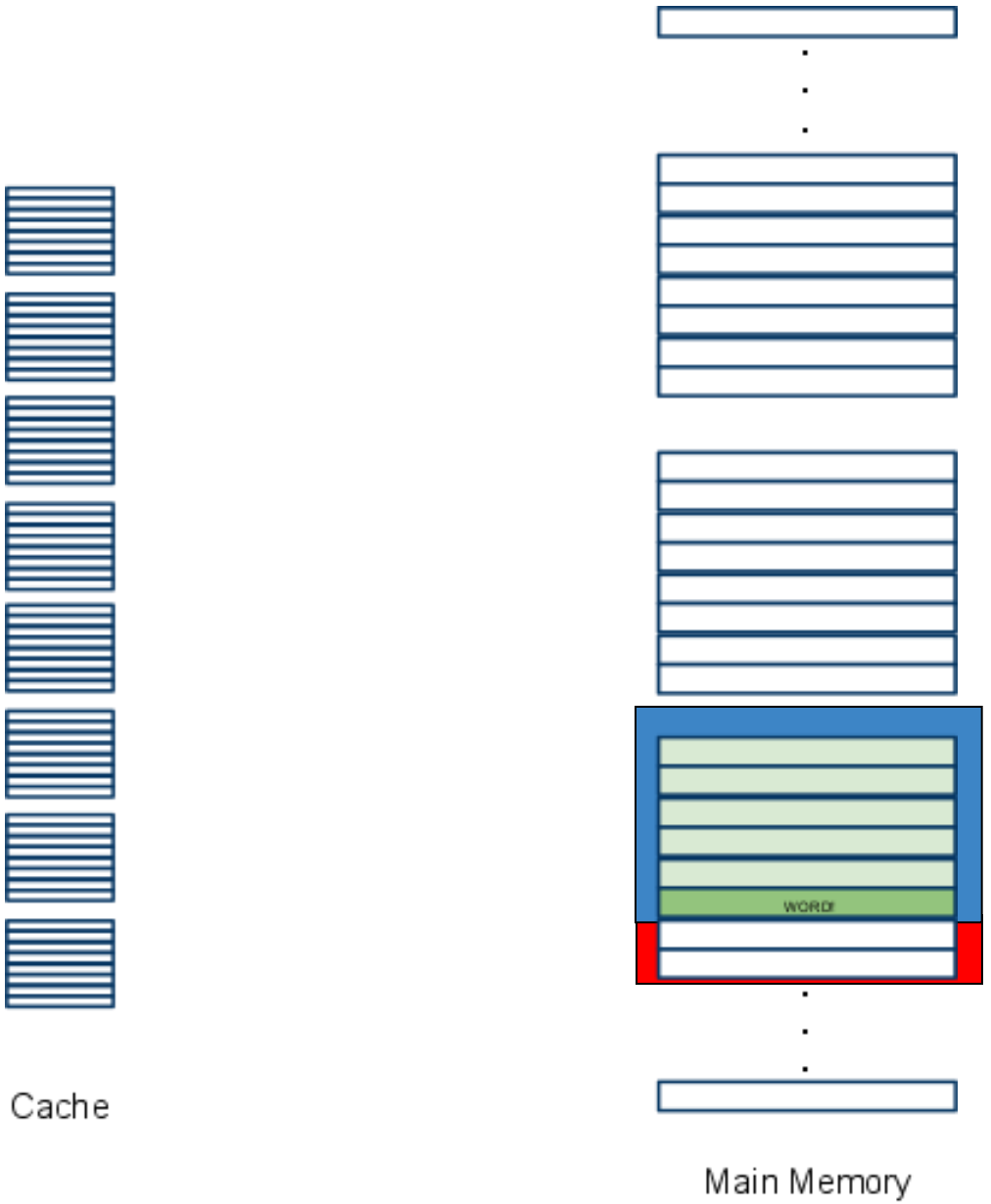
MISS!

Cache

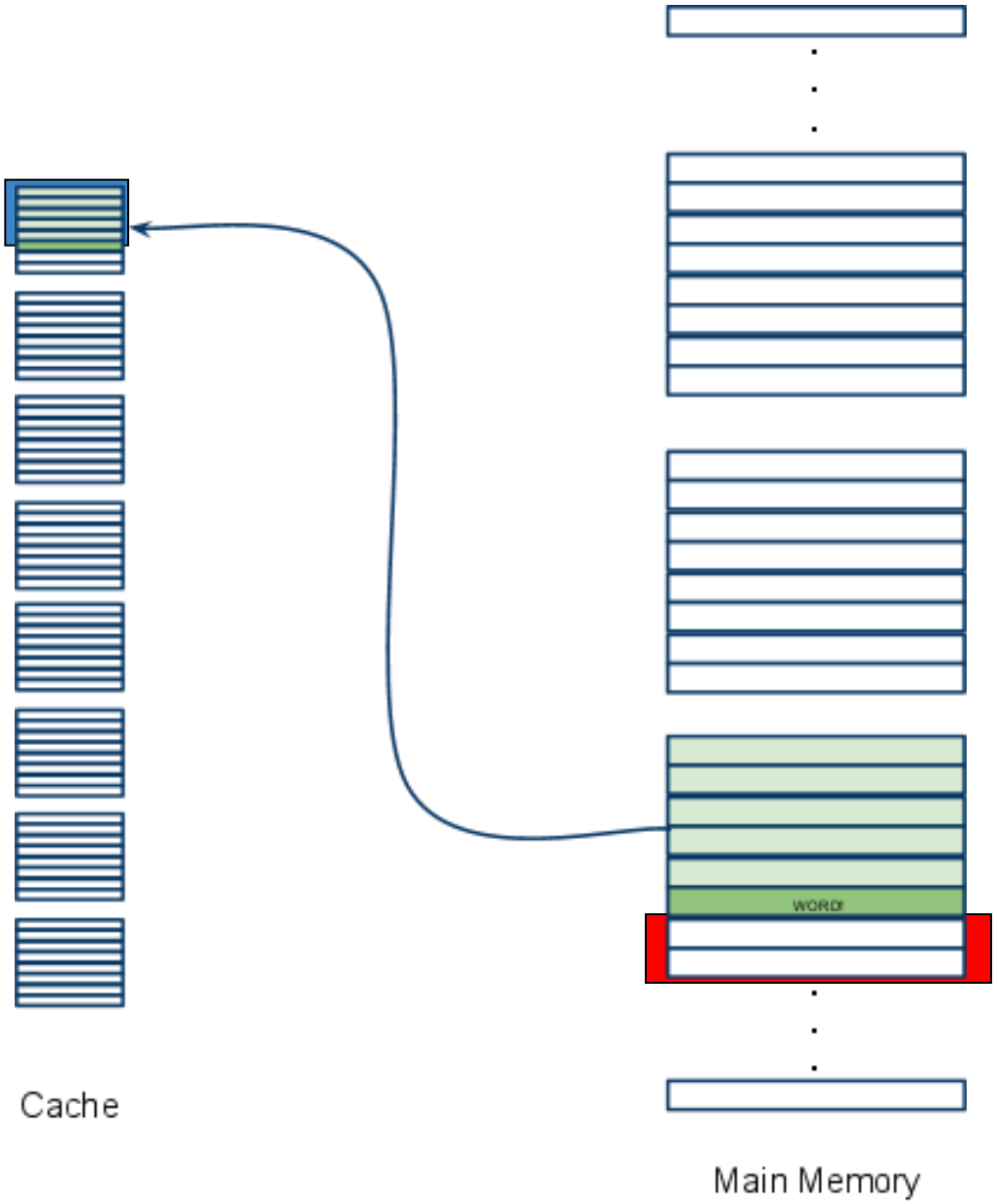


Main Memory

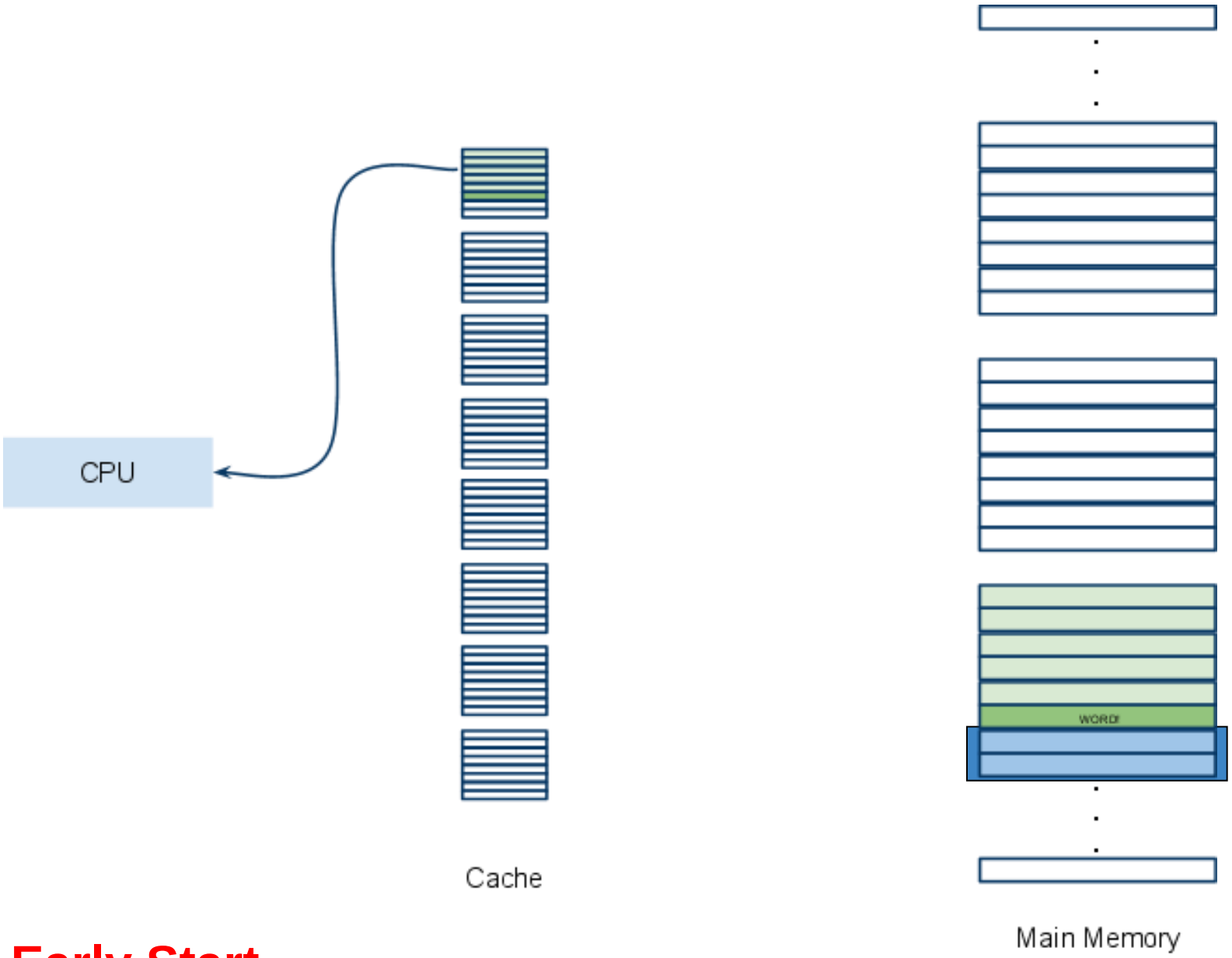
Early Start



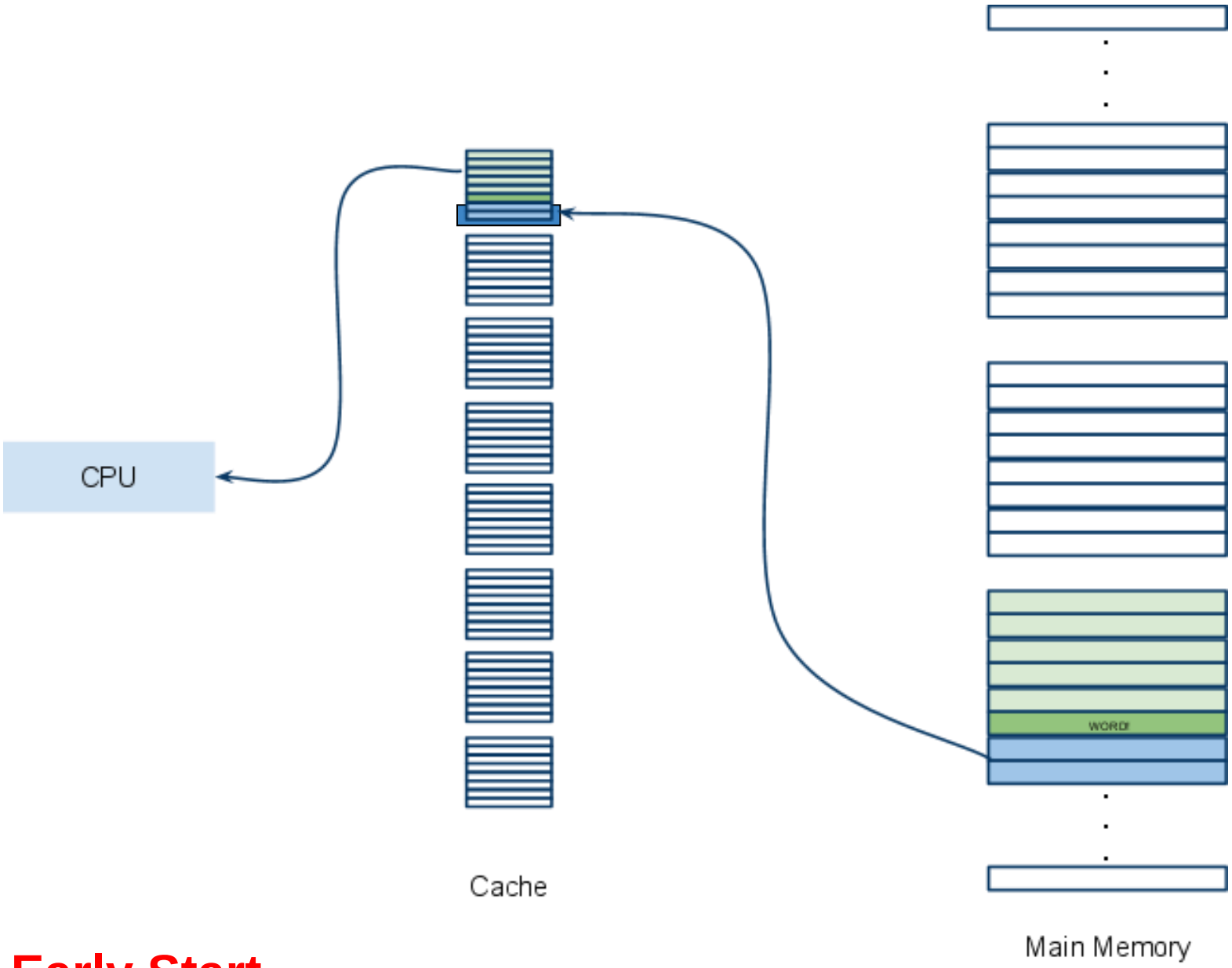
Early Start



Early Start



Early Start



Early Start

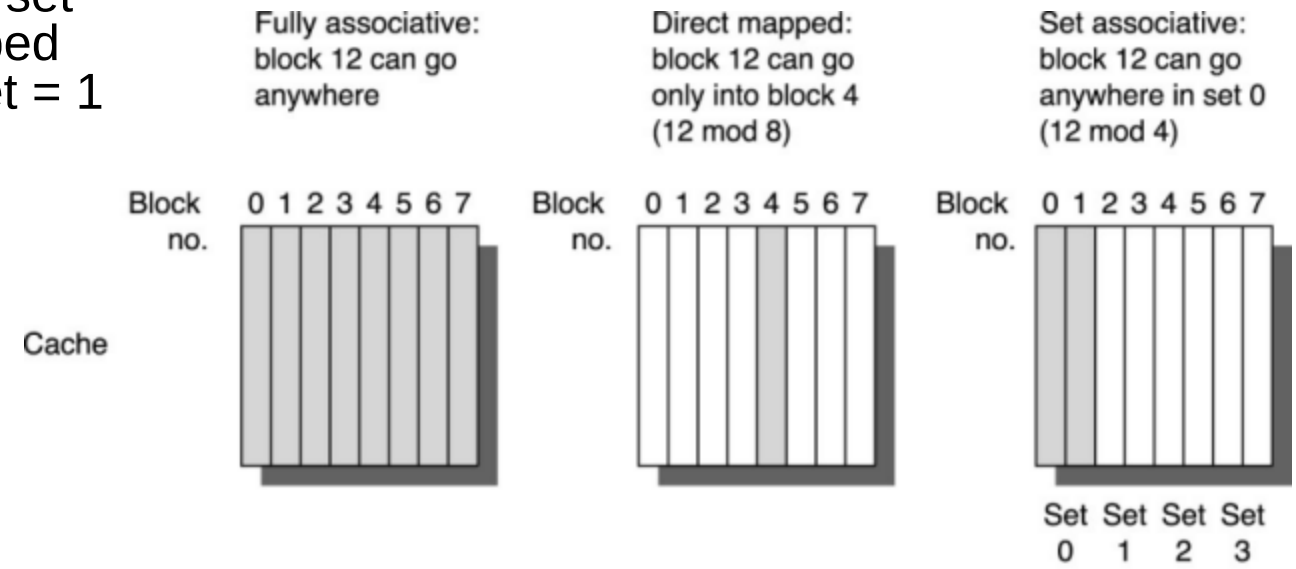
Mapping functions

- Mapping functions determine how memory blocks are placed in the cache.
- A simple processor example:
  - Cache consisting of 128 blocks of 16 words each.
  - Total size of cache is 2048 (2K) words.
  - Main memory is addressable by a 16-bit address.
  - Main memory has 64K words.
  - Main memory has 4K blocks of 16 words each.
- Three mapping functions:
  - Direct mapping
  - Associative mapping
  - Set-associative mapping.

Q1(block placement): Where can a block be placed?

Direct mapped: (*Block number)* mod (*Number of blocks in cache)*
Set associative: (*Block number)* mod (*Number of sets in cache*)
- # of set ≤ # of blocks
- n-way: n blocks in a set
- 1-way = direct mapped

Fully associative: # of set = 1



Example: block 12 placed in a cache with 8-blocks



Direct Mapping

Block j of main memory maps onto block j modulo 128 of the cache

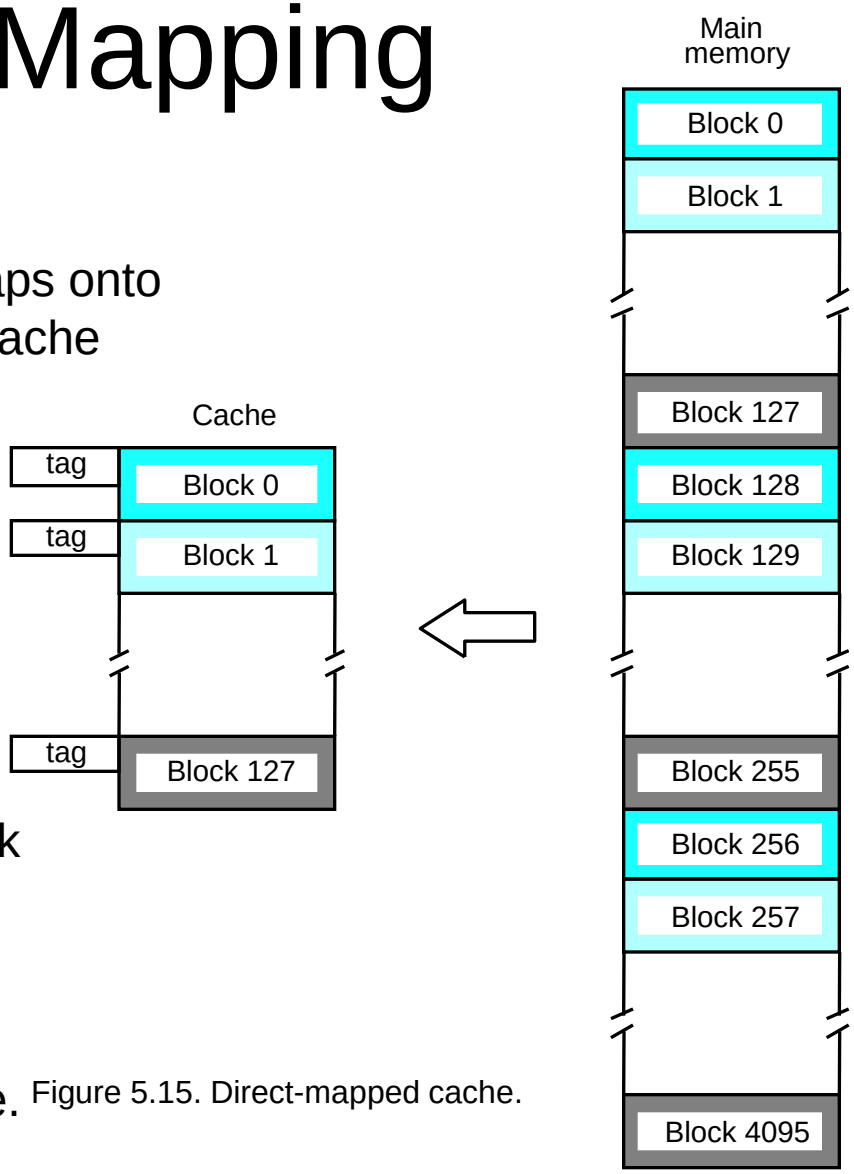
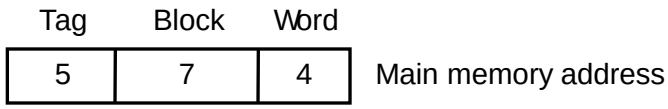


Figure 5.15. Direct-mapped cache.

4: one of 16 words. (each block has $16=2^4$ words)

7: points to a particular block in the cache ($128=2^7$)

5: 5 tag bits are compared with the tag bits associated with its location in the cache. Identify which of the 32 blocks that are resident in the cache (4096/128).



Main memory address

Direct Mapping

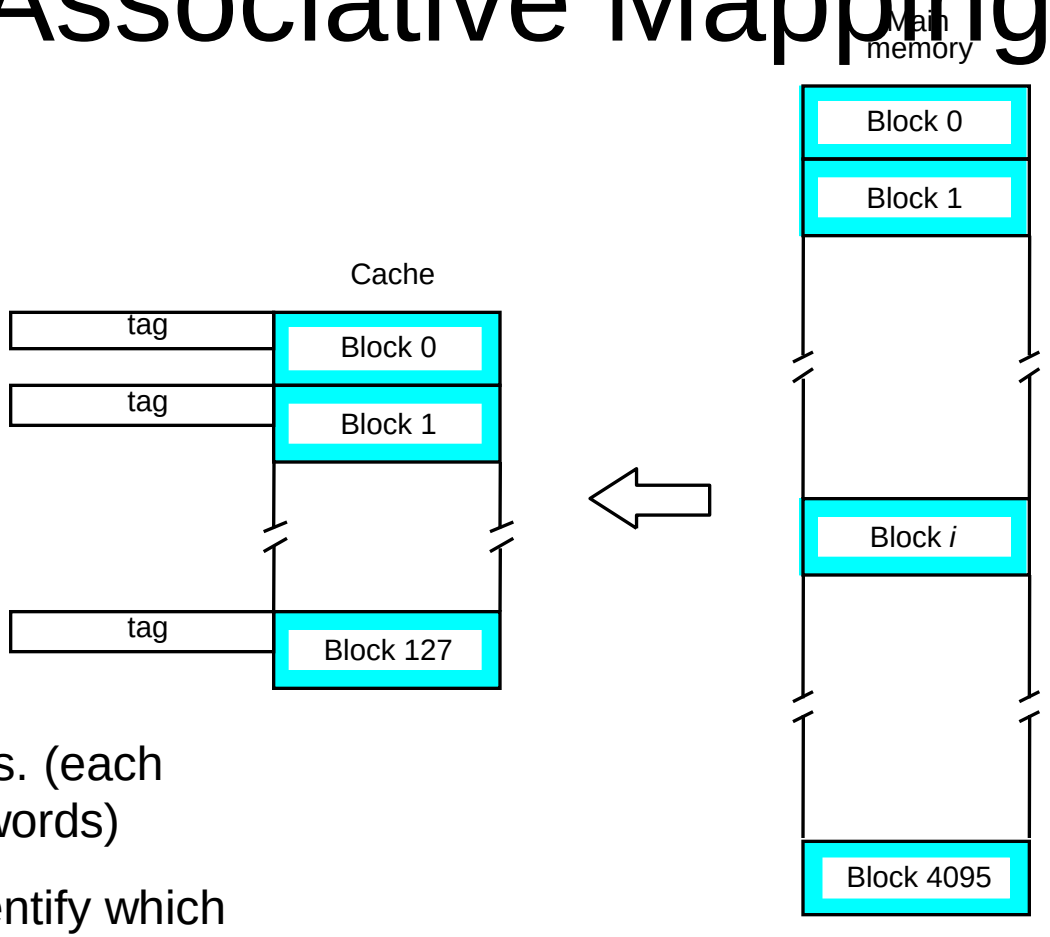
- *More than one memory block is mapped onto the same position in the cache.*
- *May lead to contention for cache blocks even if the cache is not full.*
- *Resolve the contention by allowing new block to replace the old block, leading to a trivial replacement algorithm.*
- *Memory address is divided into three fields:*

  - Low order 4 bits determine one of the 16 words in a block.

  - When a new block is brought into the cache, the next 7 bits determine which cache block this new block is placed in.

  - High order 5 bits determine which of the possible 32 blocks is currently present in the cache.
- *Simple to implement but not very flexible.*

Associative Mapping



4: one of 16 words. (each block has $16=2^4$ words)

12: 12 tag bits Identify which of the 4096 blocks that are resident in the cache $4096=2^{12}$.

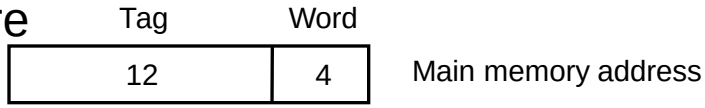


Figure 5.16. Associative-mapped cache.

Associative Mapping

- *Flexible, and uses cache space efficiently.*
- *Replacement algorithms can be used to replace an existing block in the cache when the cache is full.*
- *Cost is higher than direct-mapped cache because of the need to search all 128 patterns to determine whether a given block is in the cache.*

Set-Associative Mapping

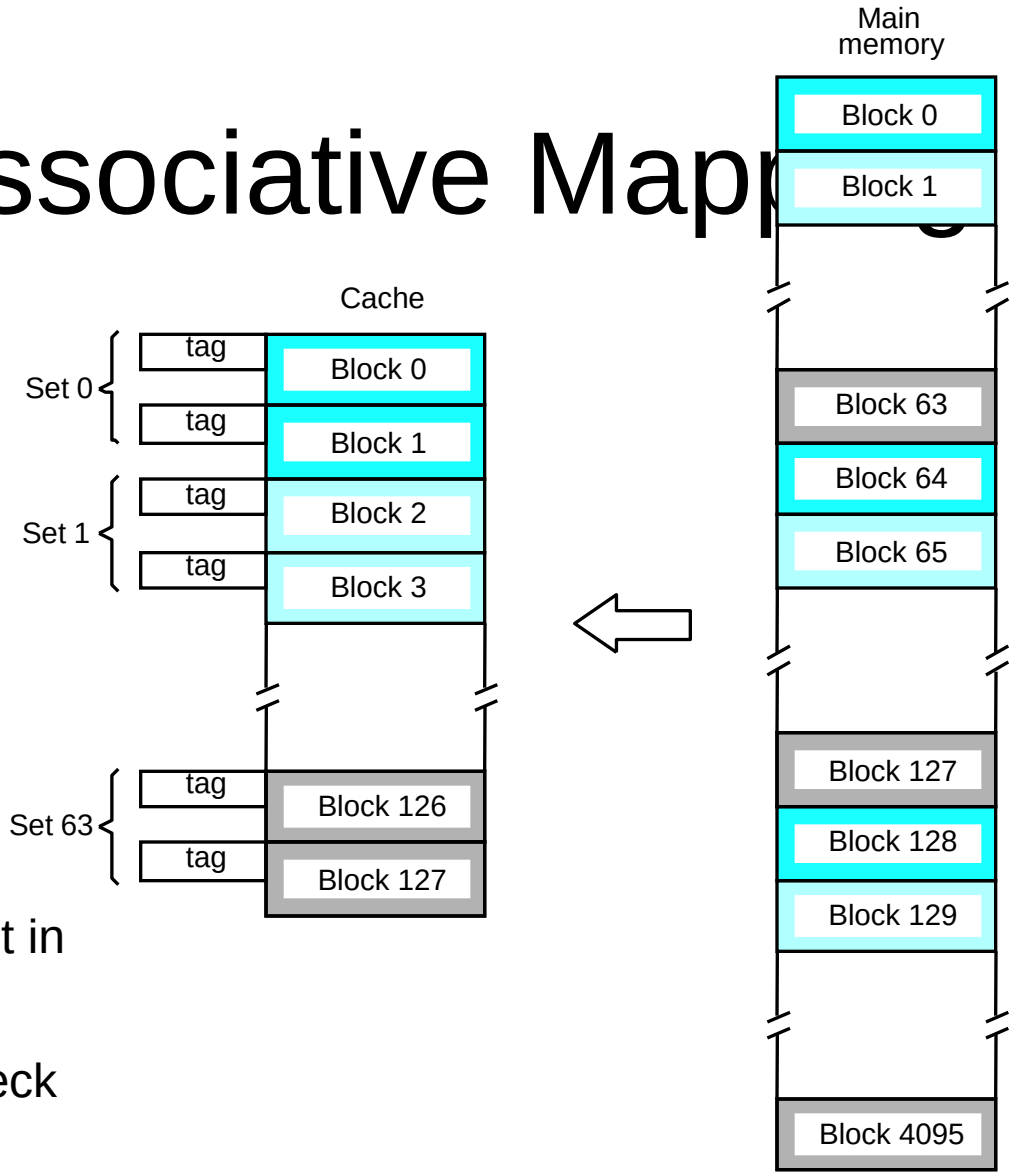
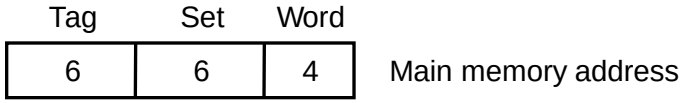


Figure 5.17. Set-associative-mapped cache with two blocks per set.

4: one of 16 words. (each block has $16=2^4$ words)

6: points to a particular set in the cache ($128/2=64=2^6$)

6: 6 tag bits is used to check if the desired block is present ($4096/64=2^6$).



Set-Associative

- *Blocks of cache are grouped into sets.*
- *Mapping function allows a block of the main memory to reside in any block of a specific set.*
- *Set-associative mapping combination of direct and associative mapping.*
- *Number of blocks per set is a design parameter.*
  - *One extreme is to have all the blocks in one set, requiring no set bits (fully associative mapping).*
  - *Other extreme is to have one block per set, is the same as direct mapping.*

Replacement Algorithms

- Difficult to determine which blocks to kick out
- Least Recently Used (LRU) block
- The cache controller tracks references to all blocks as computation proceeds.
- Increase / clear track counters when a hit/miss occurs

Replacement Algorithms

- For Associative & Set-Associative Cache

  Which location should be emptied when the cache is full and a miss occurs?

  – First In First Out (FIFO)

  – Least Recently Used (LRU)

- Distinguish an *Empty* location from a *Full* one

  – Valid Bit

Virtual Memories

Virtual memories

- Recall that an important challenge in the design of a computer system is to provide a large, fast memory system at an affordable cost.
- Architectural solutions to increase the effective speed and size of the memory system.
- Cache memories were developed to increase the effective speed of the memory system.
- Virtual memory is an architectural solution to increase the effective size of the memory system.

Virtual memories (contd..)

- Recall that the addressable memory space depends on the number of address bits in a computer.
  - For example, if a computer issues 32-bit addresses, the addressable memory space is 4G bytes.
- Physical main memory in a computer is generally not as large as the entire possible addressable space.
  - Physical memory typically ranges from a few hundred megabytes to 1G bytes.
- Large programs that cannot fit completely into the main memory have their parts stored on secondary storage devices such as magnetic disks.
  - Pieces of programs must be transferred to the main memory from secondary storage before they can be executed.

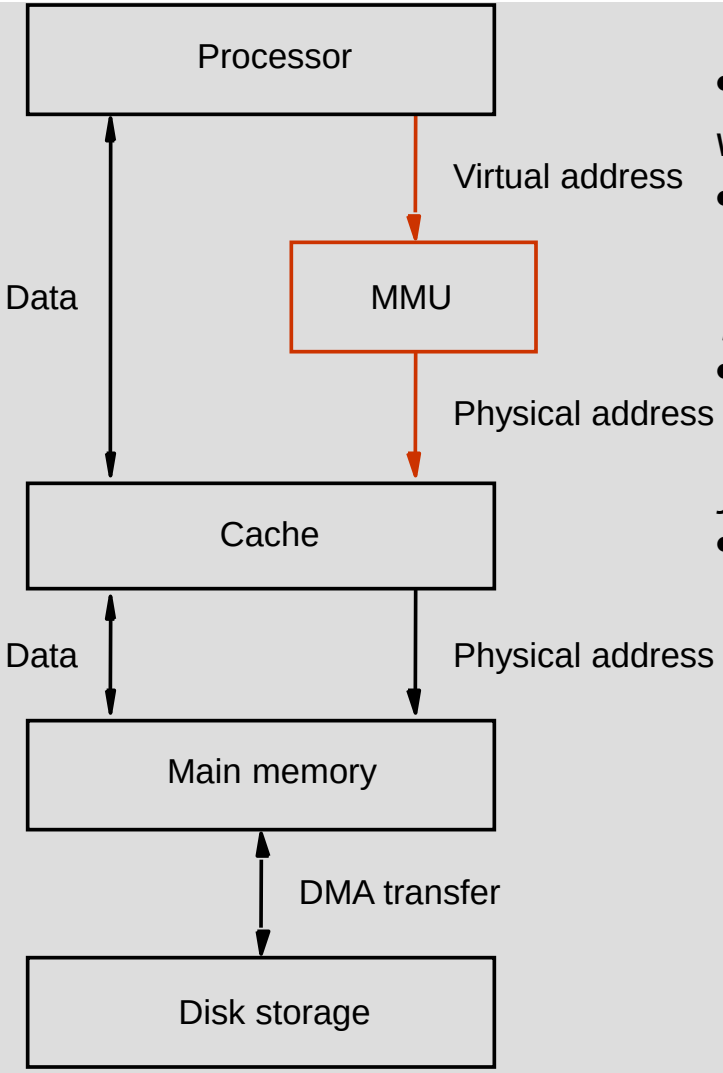
Virtual memories (contd..)

- When a new piece of a program is to be transferred to the main memory, and the main memory is full, then some other piece in the main memory must be replaced.
  - Recall this is very similar to what we studied in case of cache memories.
- Operating system automatically transfers data between the main memory and secondary storage.
  - Application programmer need not be concerned with this transfer.
  - Also, application programmer does not need to be aware of the limitations imposed by the available physical memory.

Virtual memories (contd..)

- Techniques that automatically move program and data between main memory and secondary storage when they are required for execution are called virtual-memory techniques.
- Programs and processors reference an instruction or data independent of the size of the main memory.
- Processor issues binary addresses for instructions and data.
  - These binary addresses are called logical or virtual addresses.
- Virtual addresses are translated into physical addresses by a combination of hardware and software subsystems.
  - If virtual address refers to a part of the program that is currently in the main memory, it is accessed immediately.
  - If the address refers to a part of the program that is not currently in the main memory, it is first transferred to the main memory before it can be used.

Virtual memory organization



- *Memory management unit (MMU) translates virtual addresses into physical addresses.*
- *If the desired data or instructions are in the main memory they are fetched as described previously.*
- *If the desired data or instructions are not in the main memory, they must be transferred from secondary storage to the main memory.*
- *MMU causes the operating system to bring the data from the secondary storage into the main memory.*

Address translation

- Assume that program and data are composed of fixed-length units called pages.
- A page consists of a block of words that occupy contiguous locations in the main memory.
- Page is a basic unit of information that is transferred between secondary storage and main memory.
- Size of a page commonly ranges from 2K to 16K bytes.
  - Pages should not be too small, because the access time of a secondary storage device is much larger than the main memory.
  - Pages should not be too large, else a large portion of the page may not be used, and it will occupy valuable space in the main memory.

Address translation

- Concepts of virtual memory are similar to the concepts of cache memory.
- Cache memory:
  - Introduced to bridge the speed gap between the processor and the main memory.
  - Implemented in hardware.
- Virtual memory:
  - Introduced to bridge the speed gap between the main memory and secondary storage.
  - Implemented in part by software.

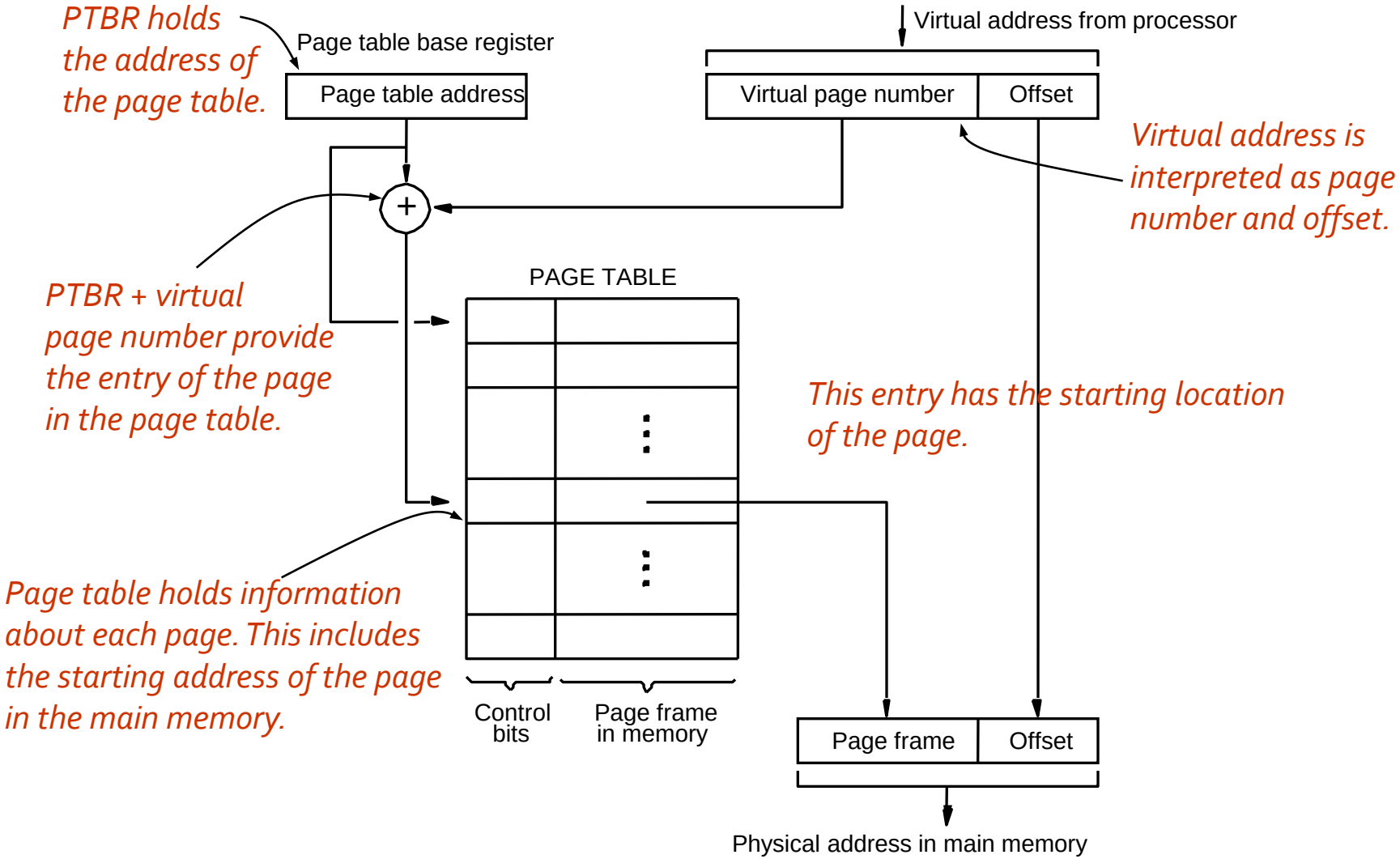
Address translation

- Each virtual or logical address generated by a processor is interpreted as a virtual page number (high-order bits) plus an offset (low-order bits) that specifies the location of a particular byte within that page.
- Information about the main memory location of each page is kept in the page table.
  - Main memory address where the page is stored.
  - Current status of the page.
- Area of the main memory that can hold a page is called as page frame.
- Starting address of the page table is kept in a page table base register.

Address translation

- Virtual page number generated by the processor is added to the contents of the page table base register.
  - This provides the address of the corresponding entry in the page table.
- The contents of this location in the page table give the starting address of the page if the page is currently in the main memory.

Address translation



Address translation

- Page table entry for a page also includes some control bits which describe the status of the page while it is in the main memory.
- One bit indicates the validity of the page.
  - Indicates whether the page is actually loaded into the main memory.
  - Allows the operating system to invalidate the page without actually removing it.
- One bit indicates whether the page has been modified during its residency in the main memory.
  - This bit determines whether the page should be written back to the disk when it is removed from the main memory.
  - Similar to the dirty or modified bit in case of cache memory.

Address translation

- Other control bits for various other types of restrictions that may be imposed.
  - For example, a program may only have read permission for a page, but not write or modify permissions.

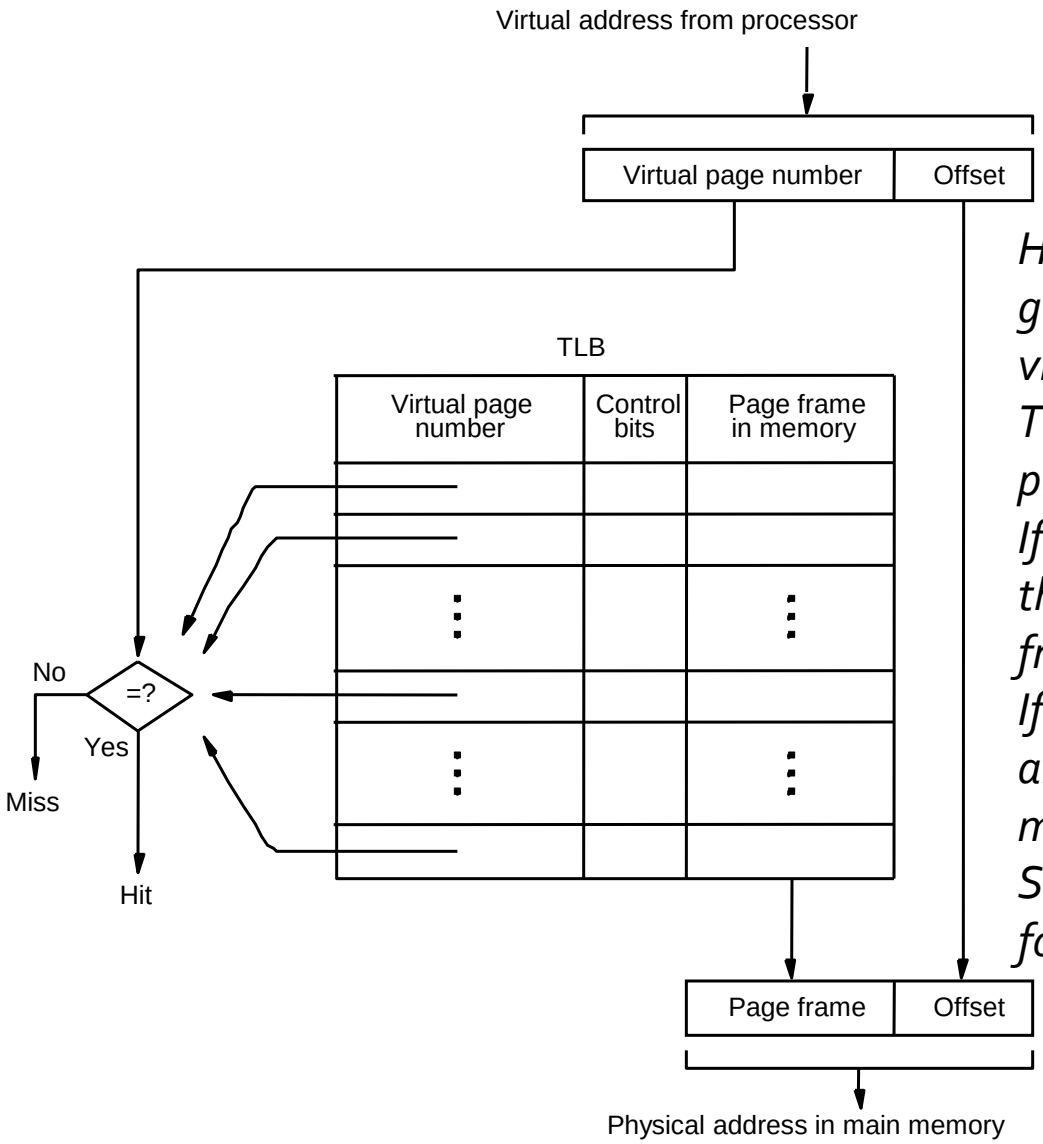
Address translation

- Where should the page table be located?
- Recall that the page table is used by the MMU for every read and write access to the memory.
  - Ideal location for the page table is within the MMU.
- Page table is quite large.
- MMU is implemented as part of the processor chip.
- Impossible to include a complete page table on the chip.
- Page table is kept in the main memory.
- A copy of a small portion of the page table can be accommodated within the MMU.
  - Portion consists of page table entries that correspond to the most recently accessed pages.

Address translation

- A small cache called as Translation Lookaside Buffer (TLB) is included in the MMU.
  - TLB holds page table entries of the most recently accessed pages.
- Recall that cache memory holds most recently accessed blocks from the main memory.
  - Operation of the TLB and page table in the main memory is similar to the operation of the cache and main memory.
- Page table entry for a page includes:
  - Address of the page frame where the page resides in the main memory.
  - Some control bits.
- In addition to the above for each page, TLB must hold the virtual page number for each page.

Address translation



Associative-mapped TLB

High-order bits of the virtual address generated by the processor select the virtual page.
These bits are compared to the virtual page numbers in the TLB.
If there is a match, a hit occurs and the corresponding address of the page frame is read.
If there is no match, a miss occurs and the page table within the main memory must be consulted.
Set-associative mapped TLBs are found in commercial processors.